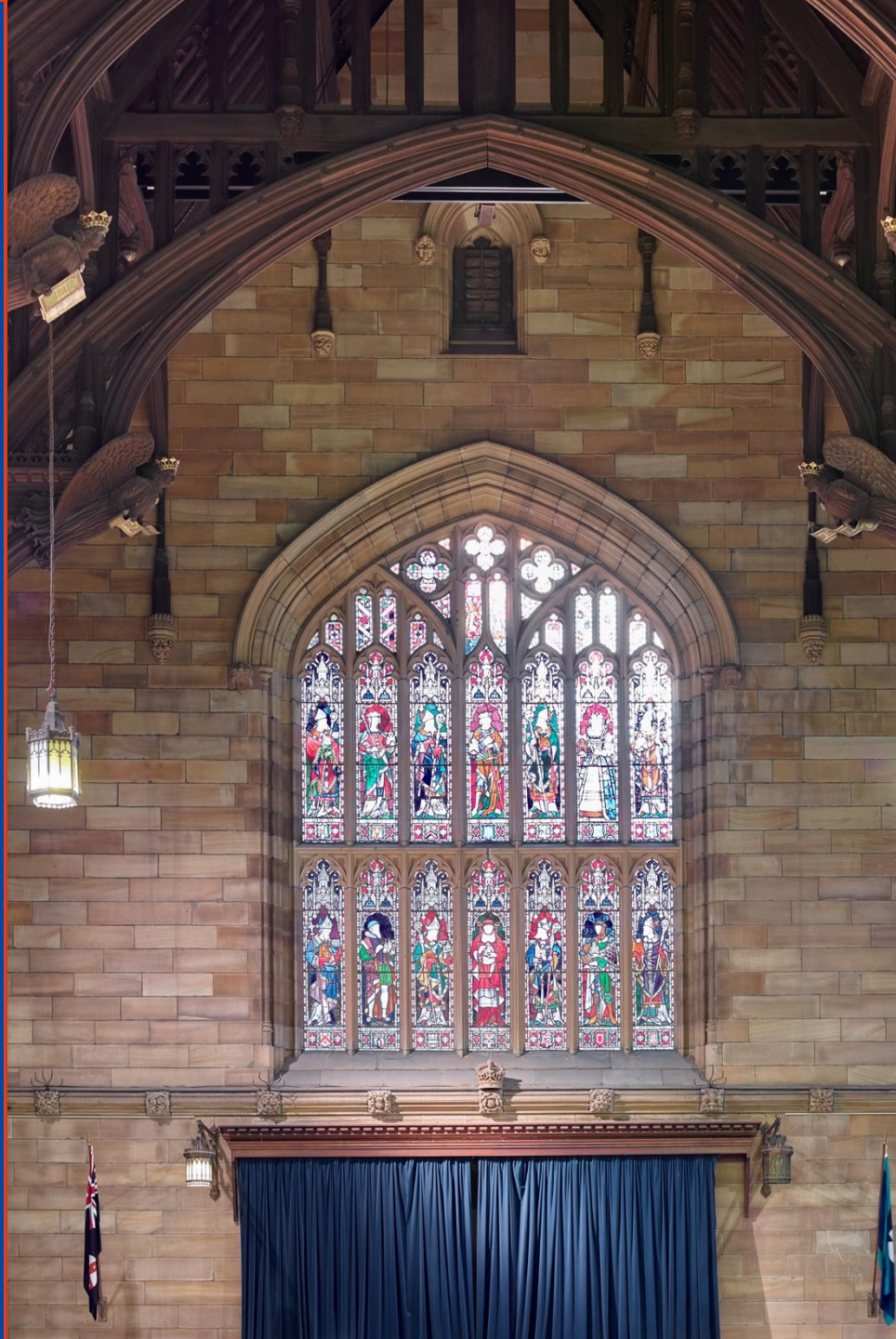
Data structures and Algorithms

Lecture 9: Graph Algorithms
[GT 14.1-2, 15.1-3]

Lecturer: Dr. Armin Chitizadeh

School of Computer Science

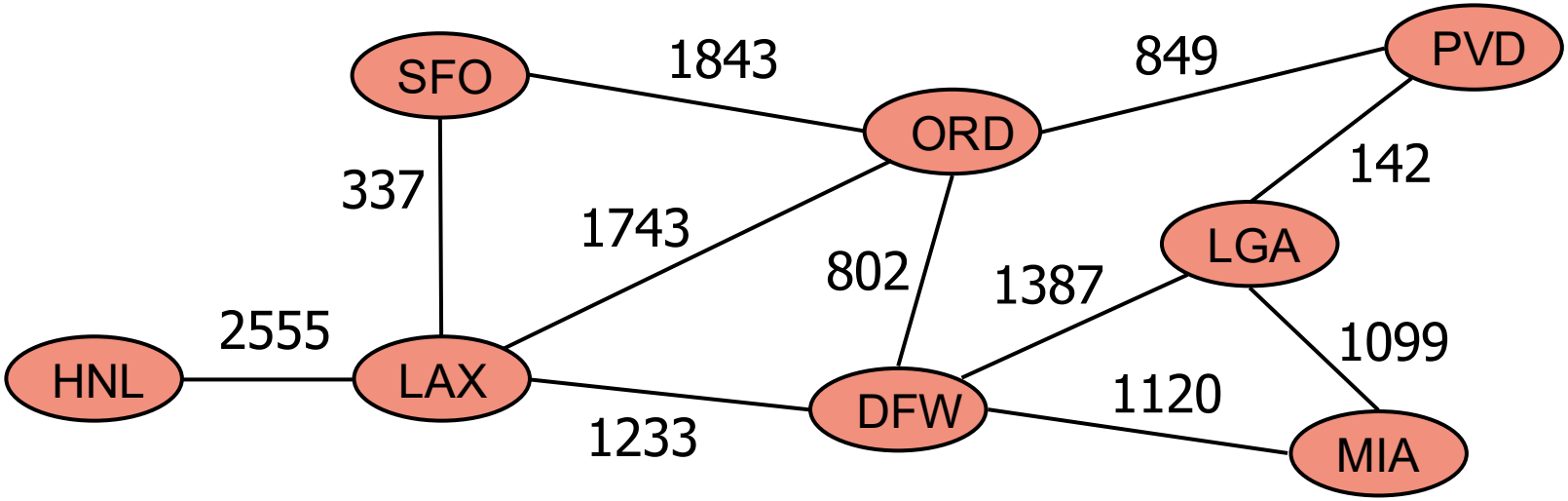
Inspired by lectures by Dr. Karlos Ishac and Dr. Andre van Renssen.



Weighted Graphs

- In a weighted graph, each edge has an associated numerical value, called the weight of the edge
- Edge weights may represent, distances, costs, etc.

Example: In a flight route graph, the weight of an edge represents the distance in miles between the endpoint airports

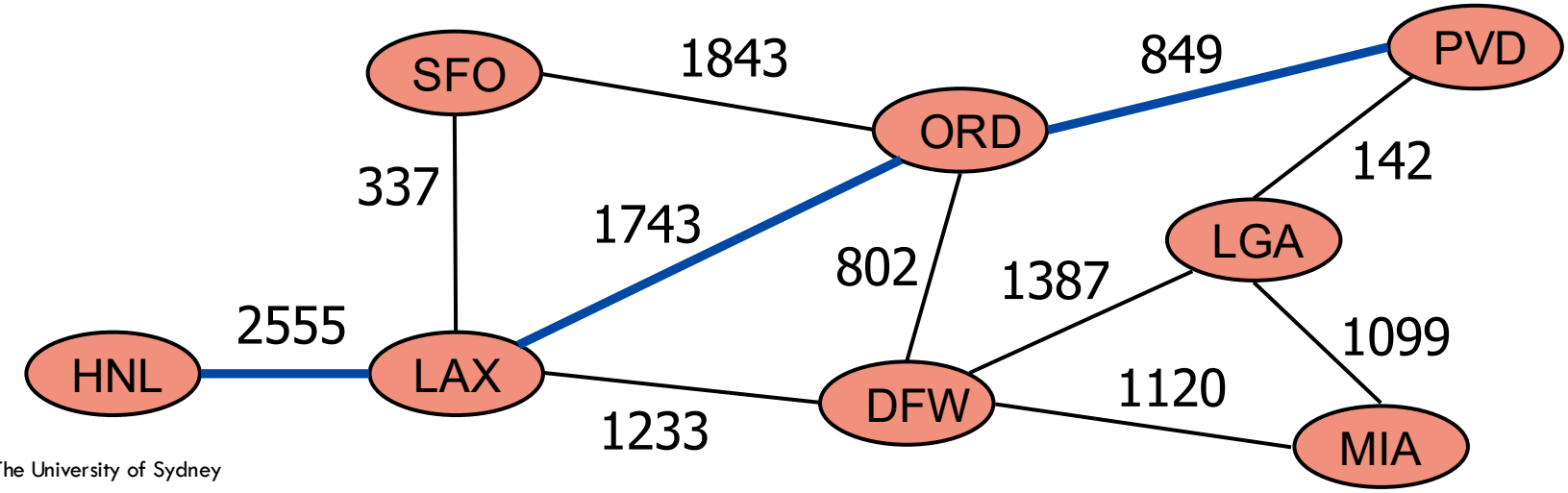


Shortest Paths

Given an edge weighted graph and two vertices u and v, we want to find a path of minimum total weight between u and v, where the weight of a path is the sum of the weights of its edges.

Applications: Internet packet routing, flight reservations and driving directions.

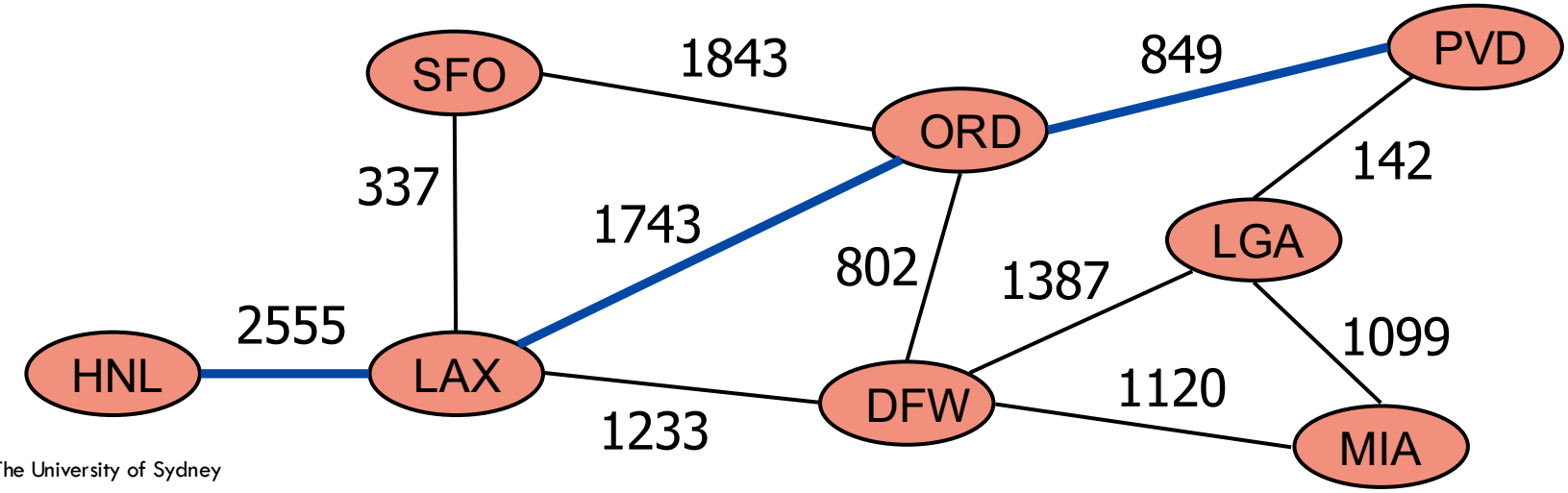
Example: Shortest path between Providence (PVD) and Honolulu (HNL)



Shortest Path Properties

Property: A subpath of a shortest path is itself a shortest path

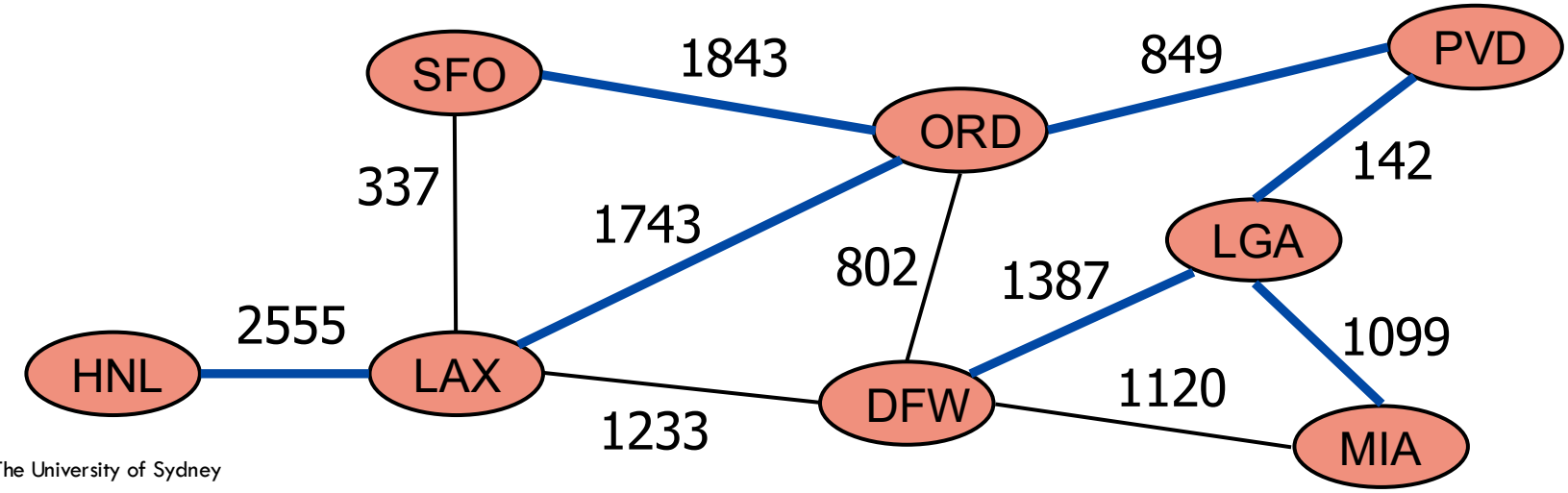
Example: Shortest path from Providence (PVD) to Honolulu (HNL) also contains a shortest path from Providence (PVD) to Los Angeles (LAX)



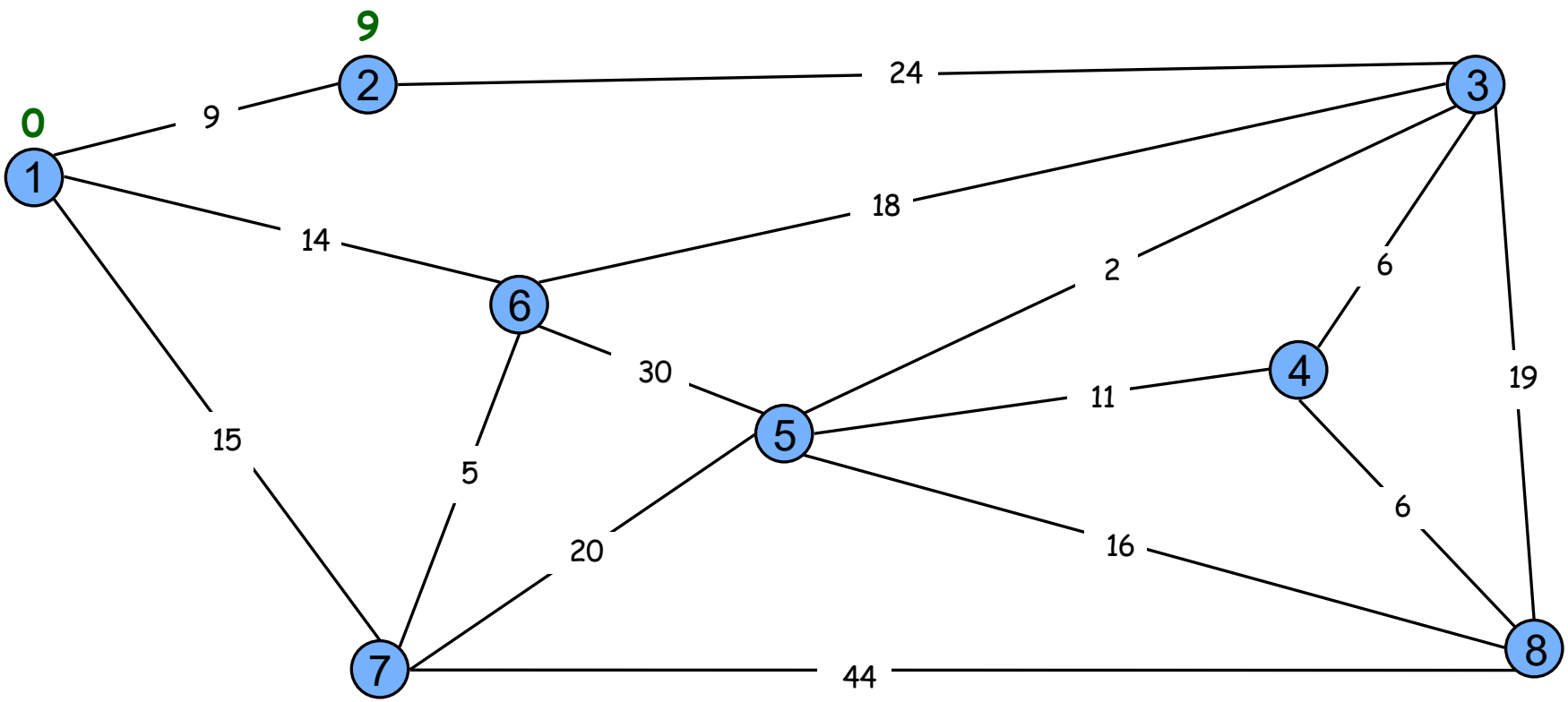
Shortest Path Properties

Property: There is a tree of shortest paths from a start vertex to all the other vertices (shortest path tree).

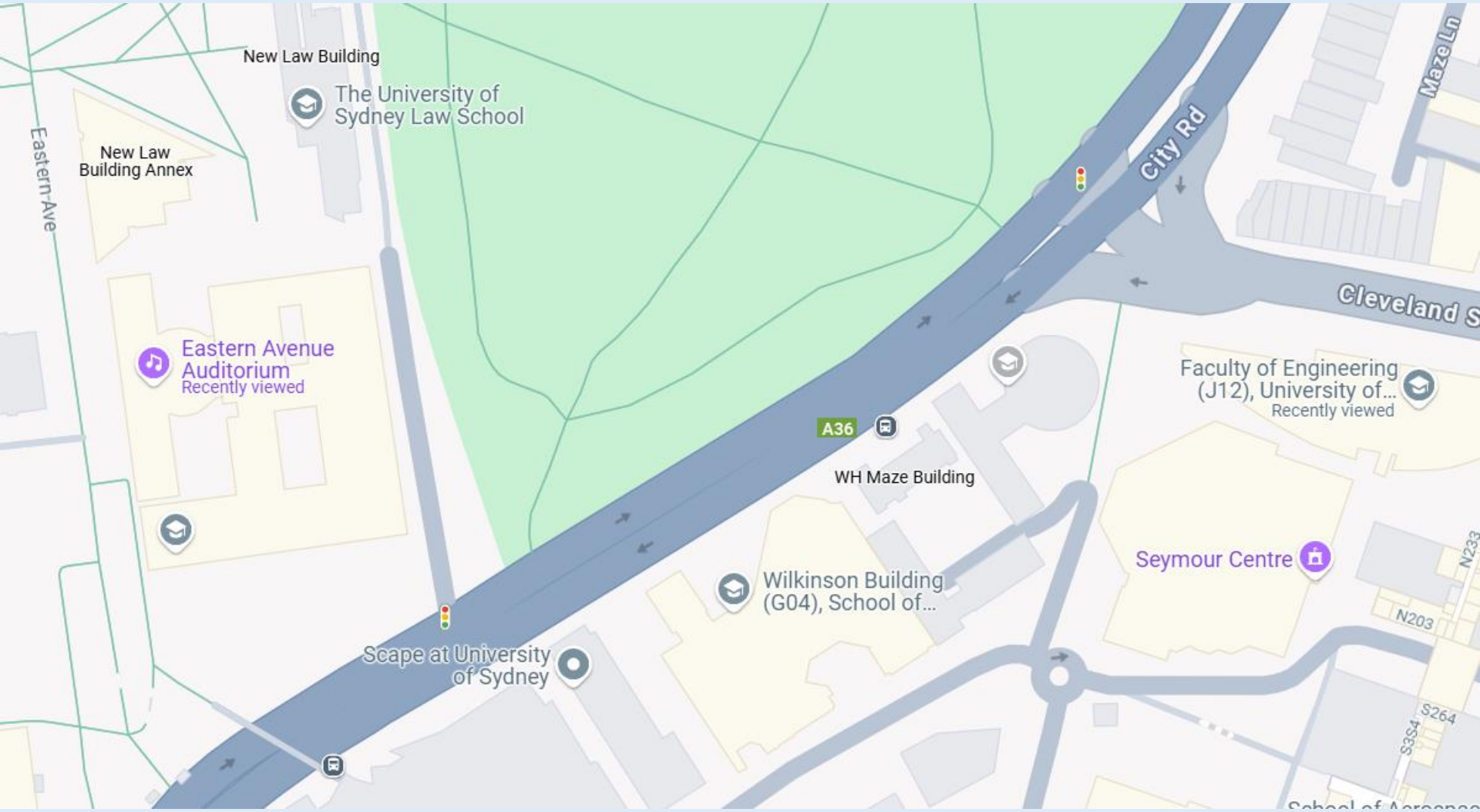
Example: Tree of shortest paths from Providence (PVD)



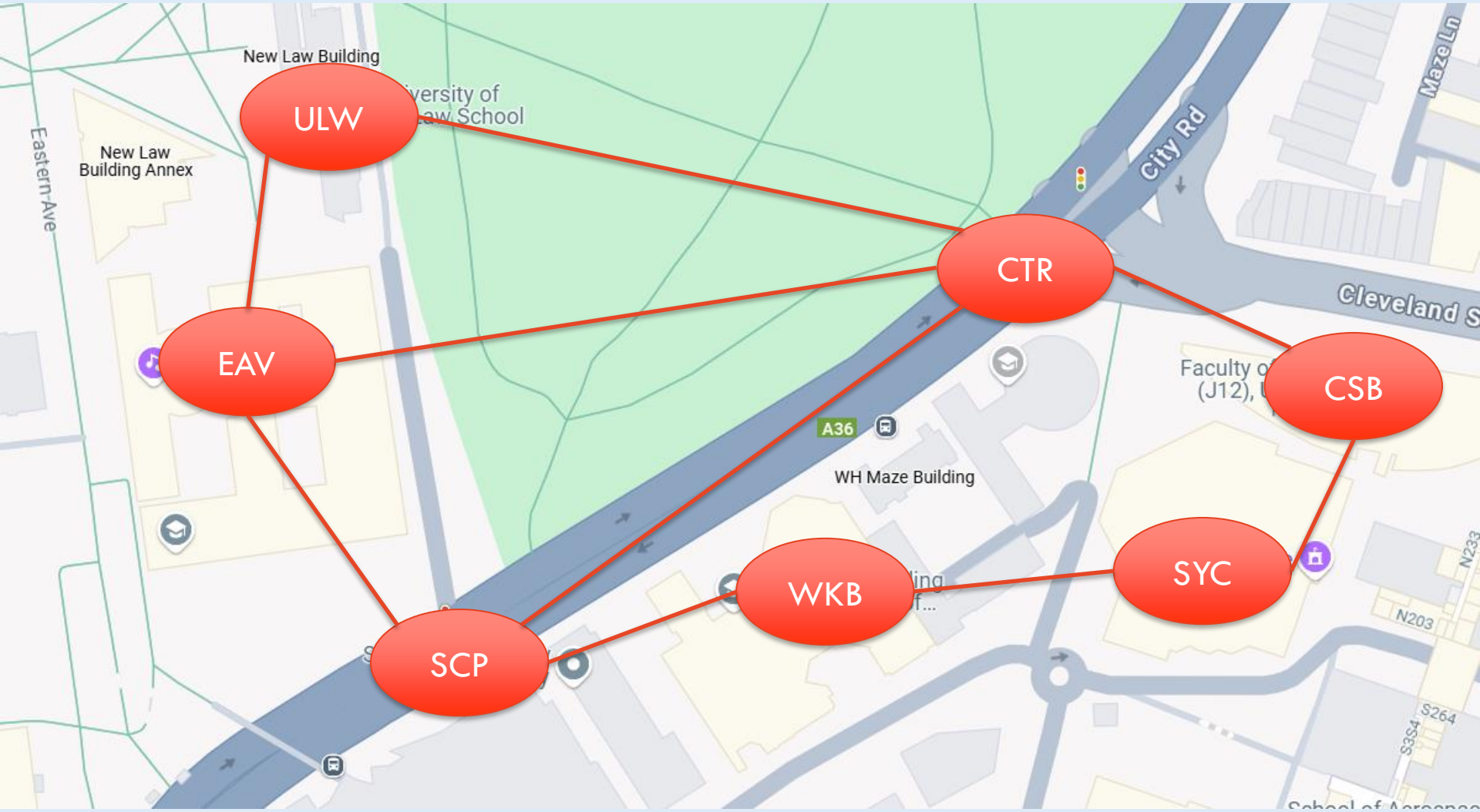
Shortest Paths



Interactive Example: Help me get from J12 to Eastern Avenue



Interactive Example: Help me get from J12 to Eastern Avenue



When do you try to find the shortest path in daily life?

Dijkstra's Algorithm

Input:

- Graph $G = (V, E)$
- Edges weights $w : E \rightarrow R_+$
- Start vertex s

Output:

- Distance from s to all v in V
- Shortest path tree rooted at s

Assumptions:

- G is connected and undirected
- edge weights are nonnegative

High level idea:

- Maintain a distance estimate
  $D[v] \geq dist_w(s, v)$ for all v in V
- Keep track of a subset S of V s.t.
  $D[v] = dist_w(s, v)$ for all v in S

Initially:

- $D[s] = 0$
- $D[v] = \infty$ for all v in $V - s$

In each iteration we:

- add to S vertex u in $V \setminus S$ with smallest $D[u]$
- update D-values for vertices adjacent to u

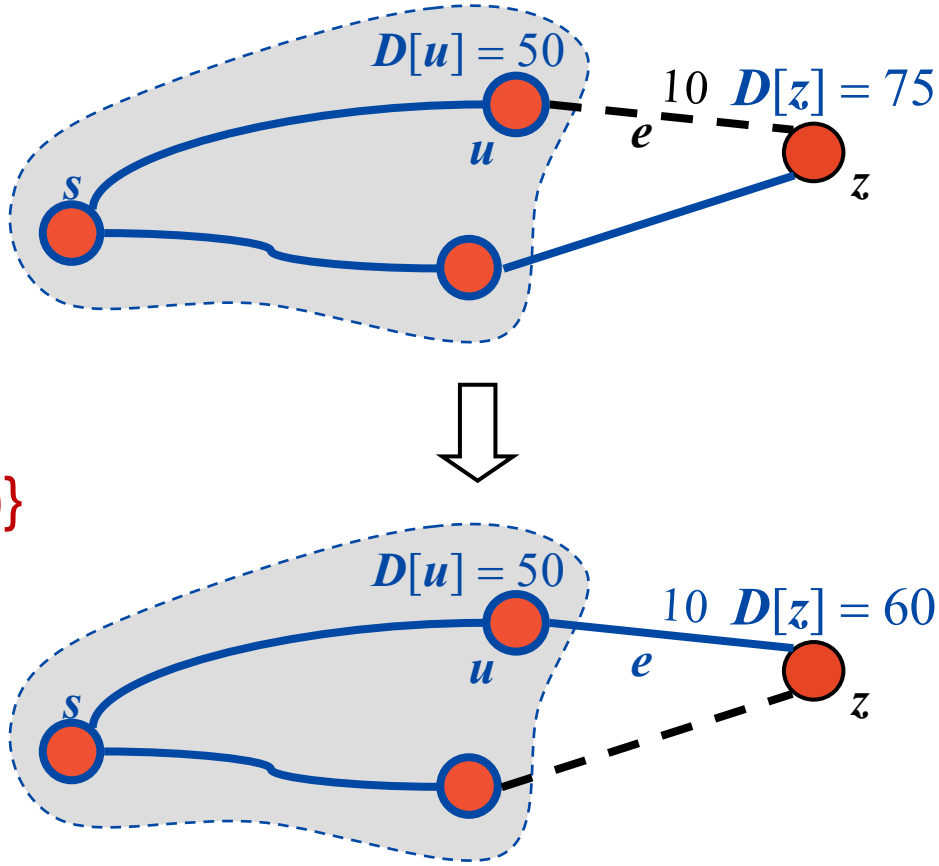
Edge Relaxation

Consider edge e = (u, z) such that:

- u is the last vertex added to S
- z is not in S

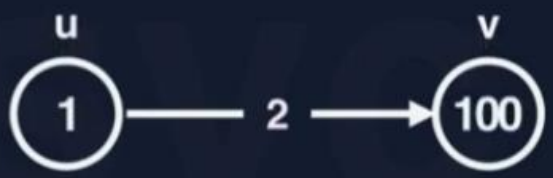
The relaxation of edge (u, z) updates D[z] as follows:

$$D[z] \leftarrow \min\{D[z], D[u] + w(u, z)\}$$



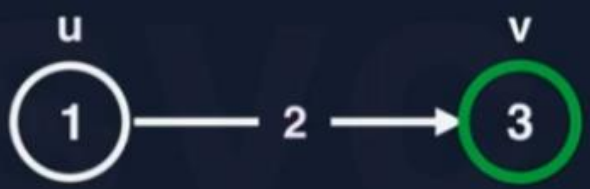
Edge Relaxation

```
RELAX(u, v, weight) {
  if v.key > u.key + weight(u, v)
    v.key = u.key + weight(u, v)
    v.previous = u
}
```



After Edge Relaxation

```
RELAX(u, v, weight) {
  if v.key > u.key + weight(u, v)
    v.key = u.key + weight(u, v)
    v.previous = u
}
```



Credit: Computer Psychology YT Channel

Dijkstra's Algorithm pseudocode

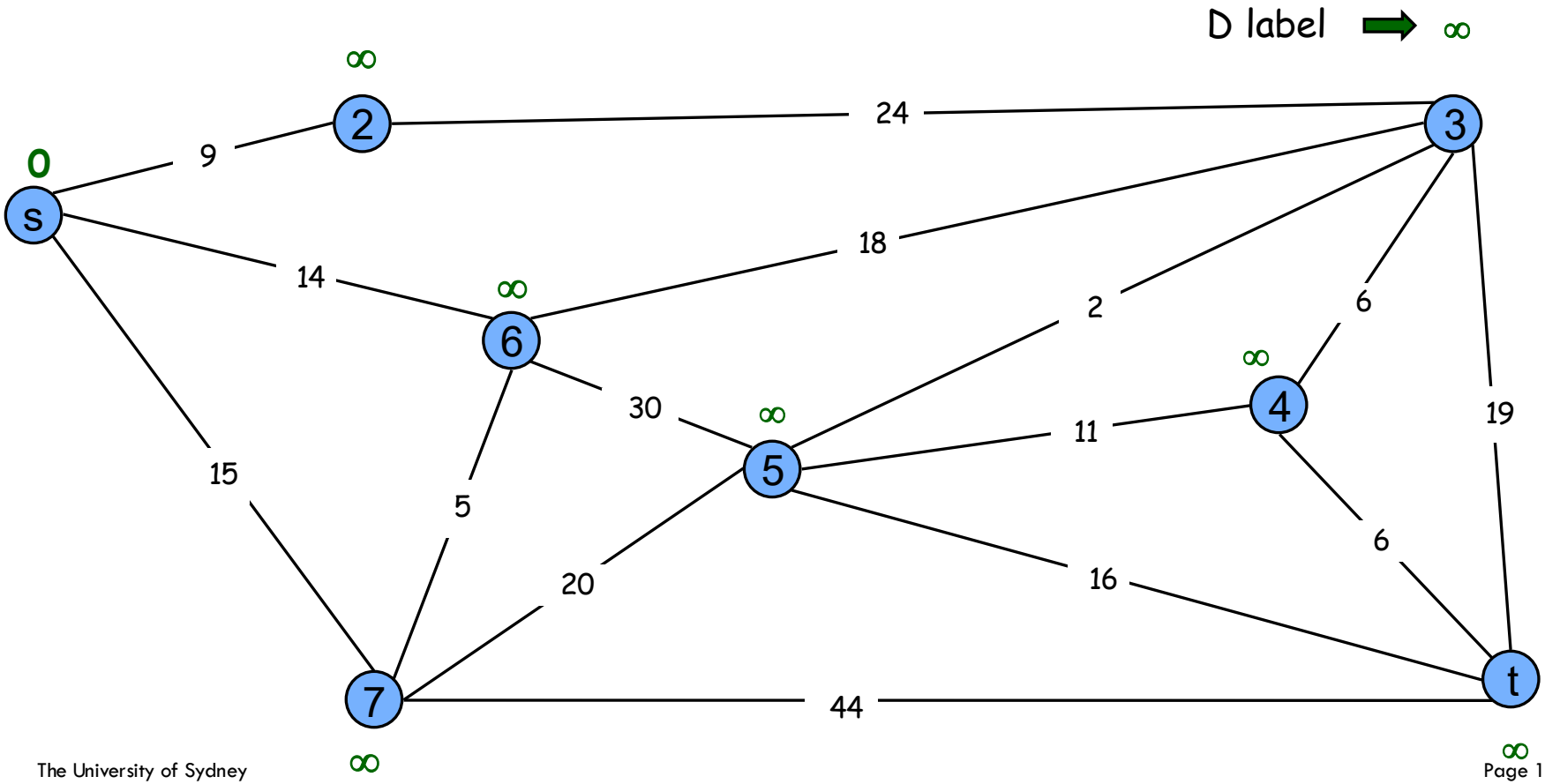
```
def Dijkstra(G, w, s):

  # initialize algorithm
  for v in V do
    D[v] ← ∞
    parent[v] ← ∅
  D[s] ← 0
  Q ← new priority queue for { (v, D[v]) : v in V }

  # iteratively add vertices to S
  while Q is not empty do
    u ← Q.remove_min()
    for z in G.neighbors(u) do
      if D[u] + w[u, z] < D[z] then
        D[z] ← D[u] + w[u, z]
        Q.update_priority(z, D[z])
        parent[z] ← u
  return D, parent
```

Dijkstra's Shortest Path Algorithm

S = { }
PQ = { s, 2, 3, 4, 5, 6, 7, t }



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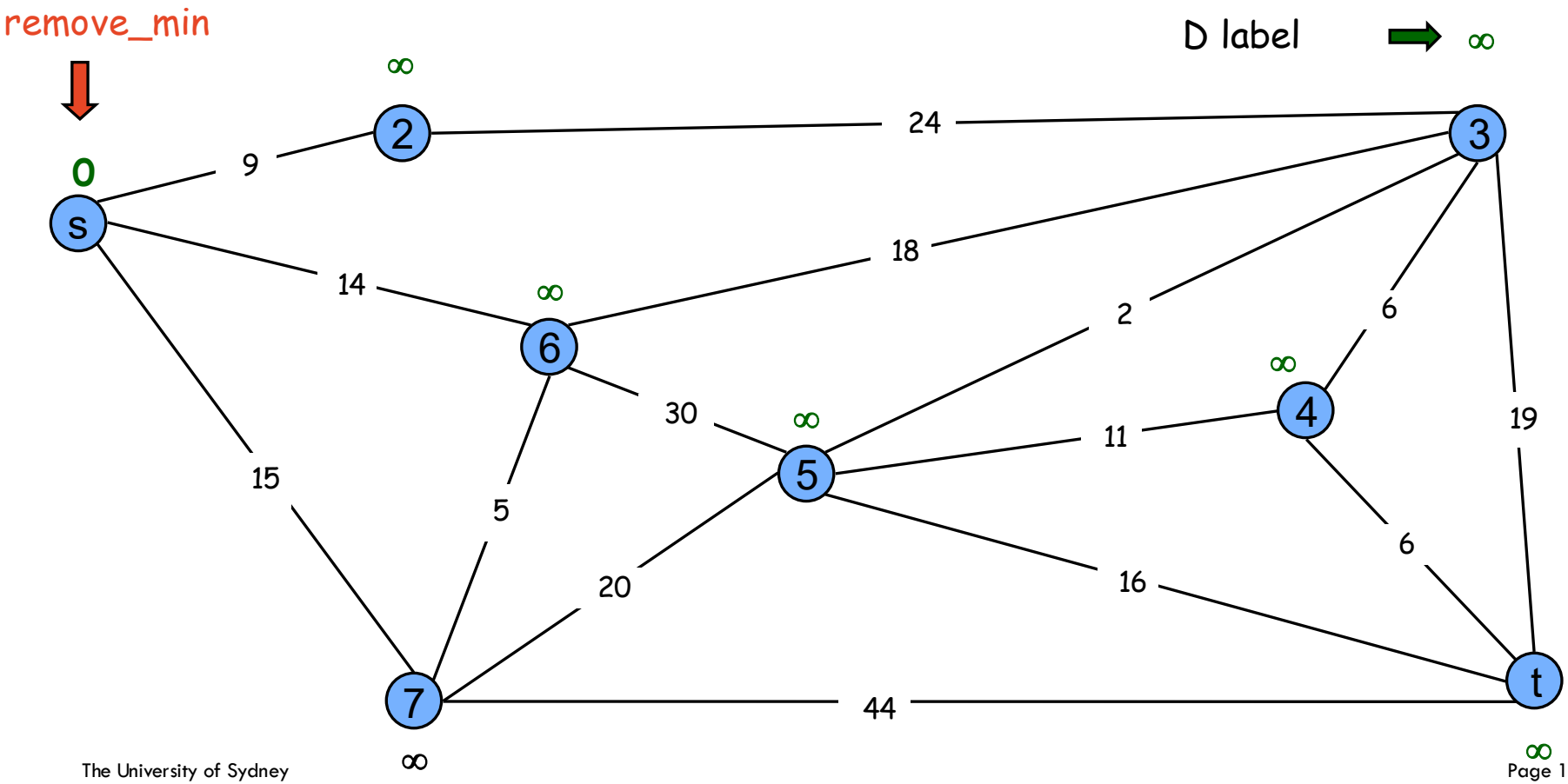

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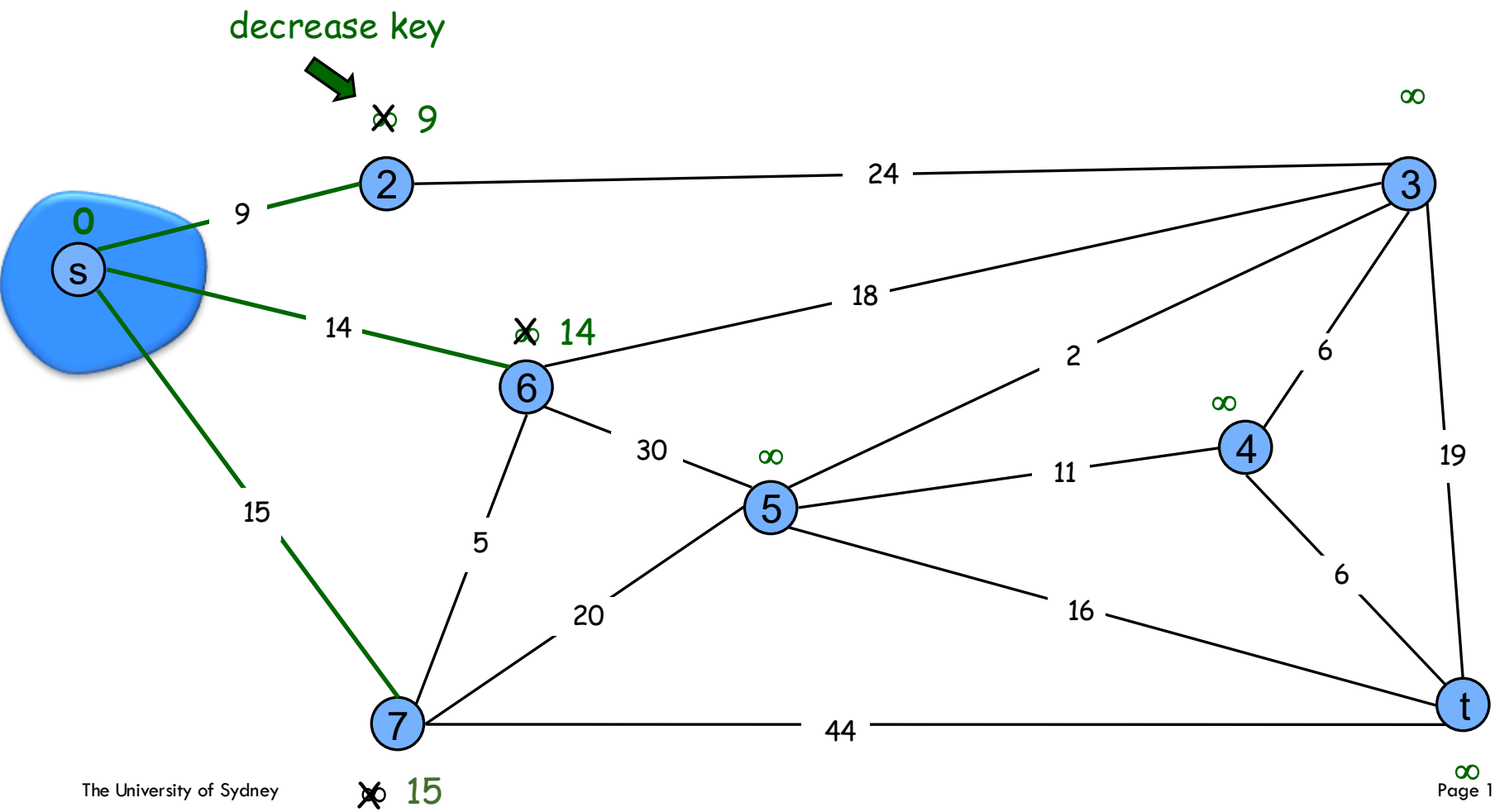

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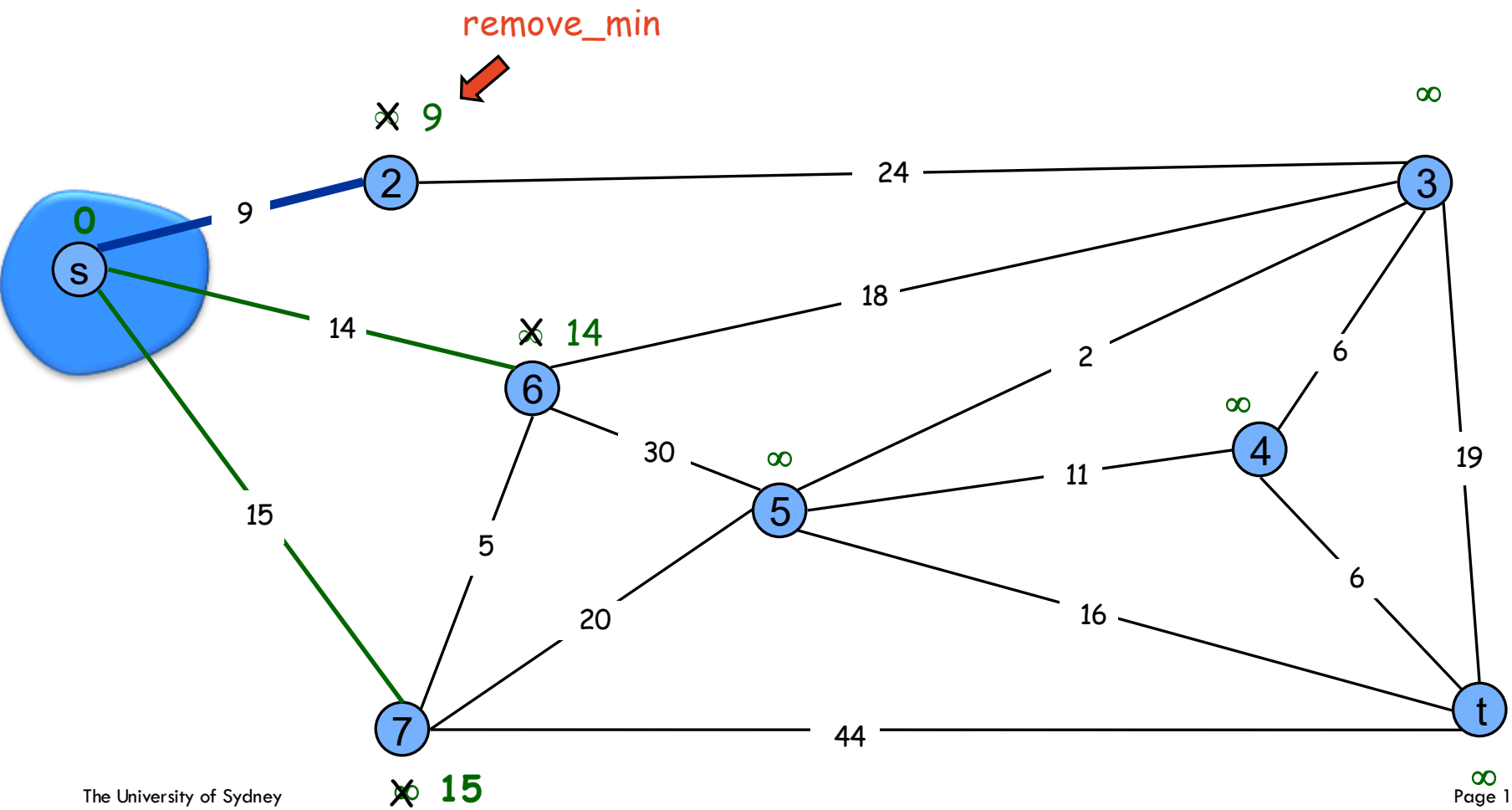
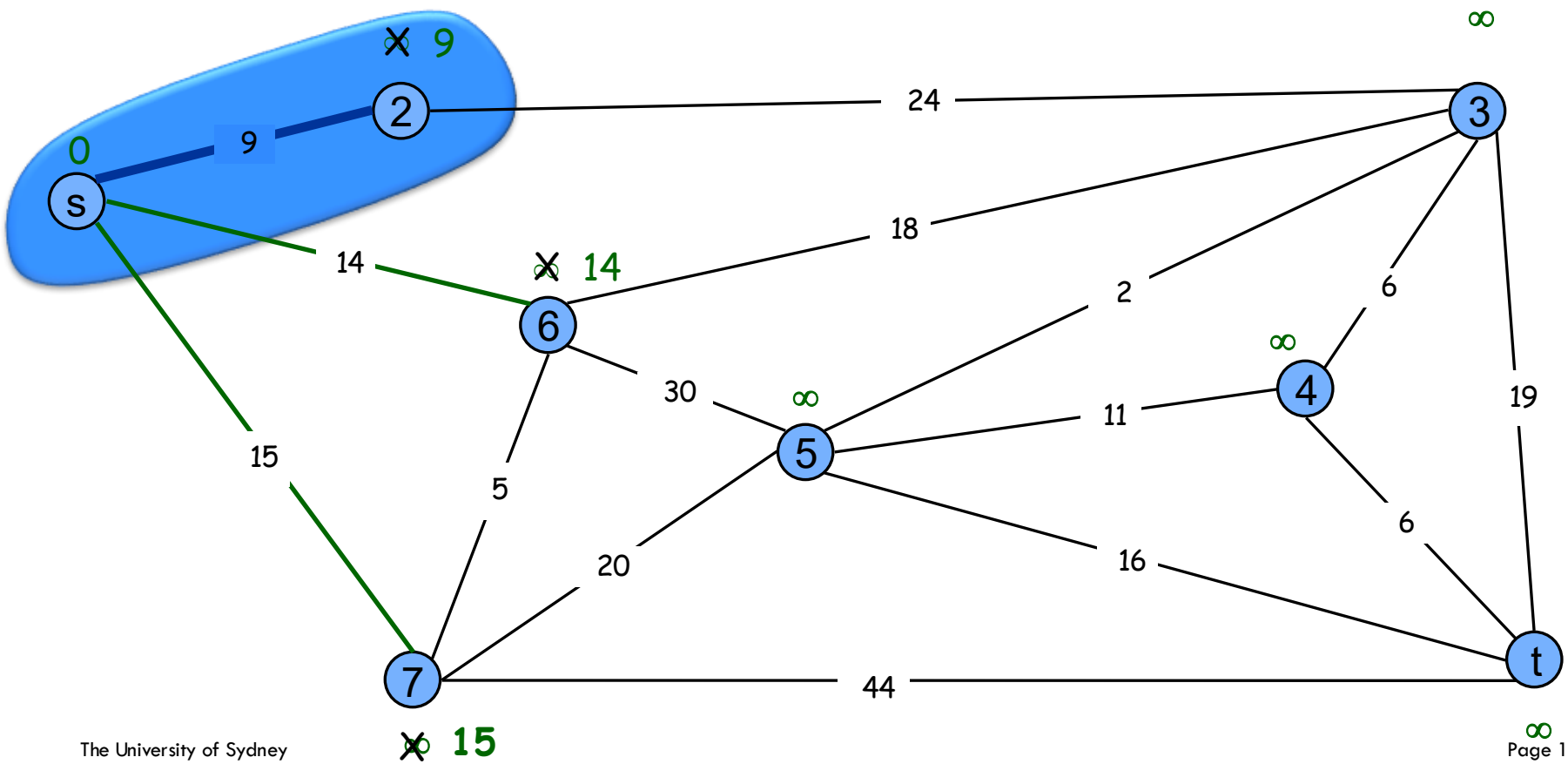


Dijkstra's Shortest Path Algorithm

S = { s, 2 }
PQ = { 3, 4, 5, 6, 7, t }



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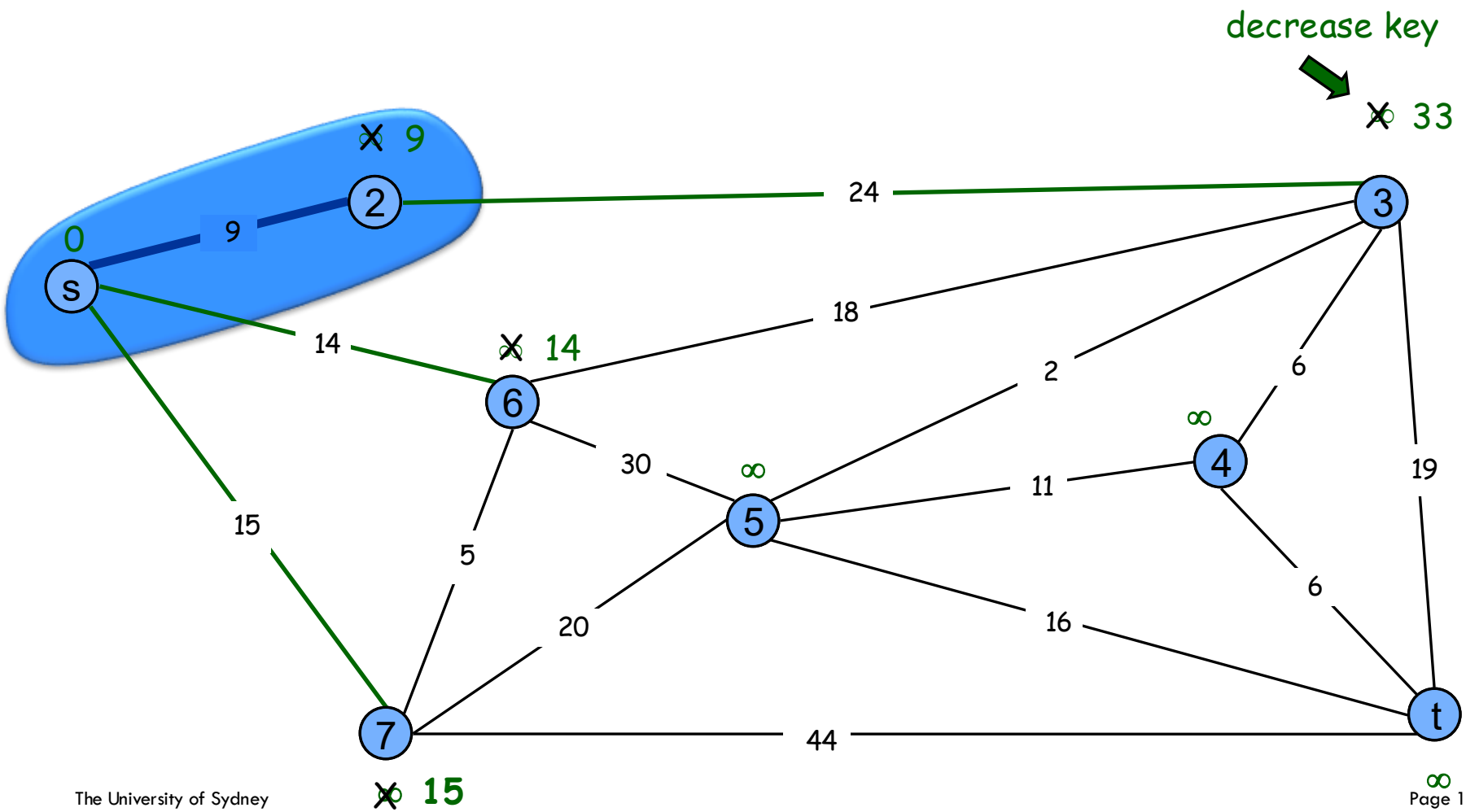

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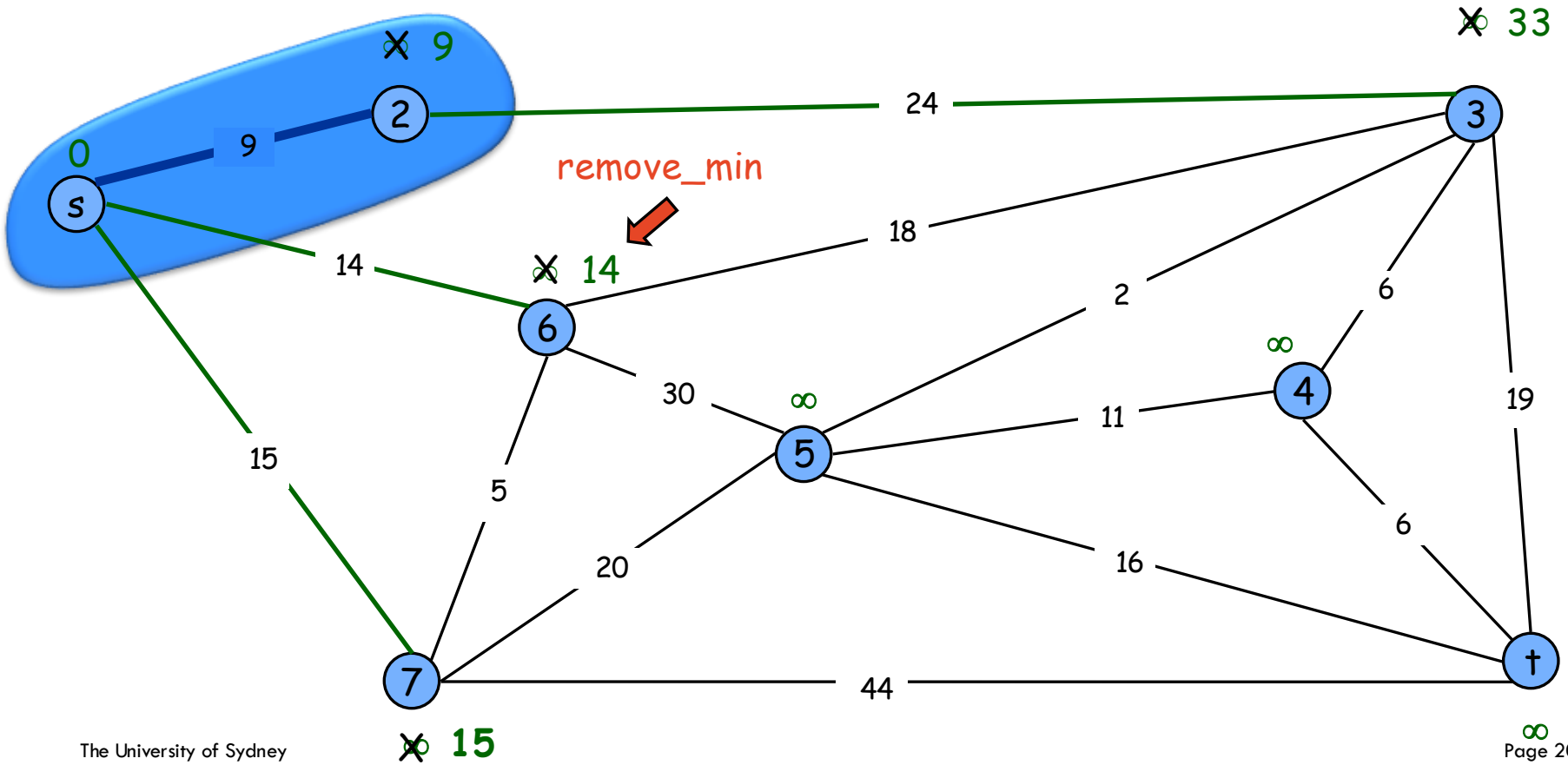
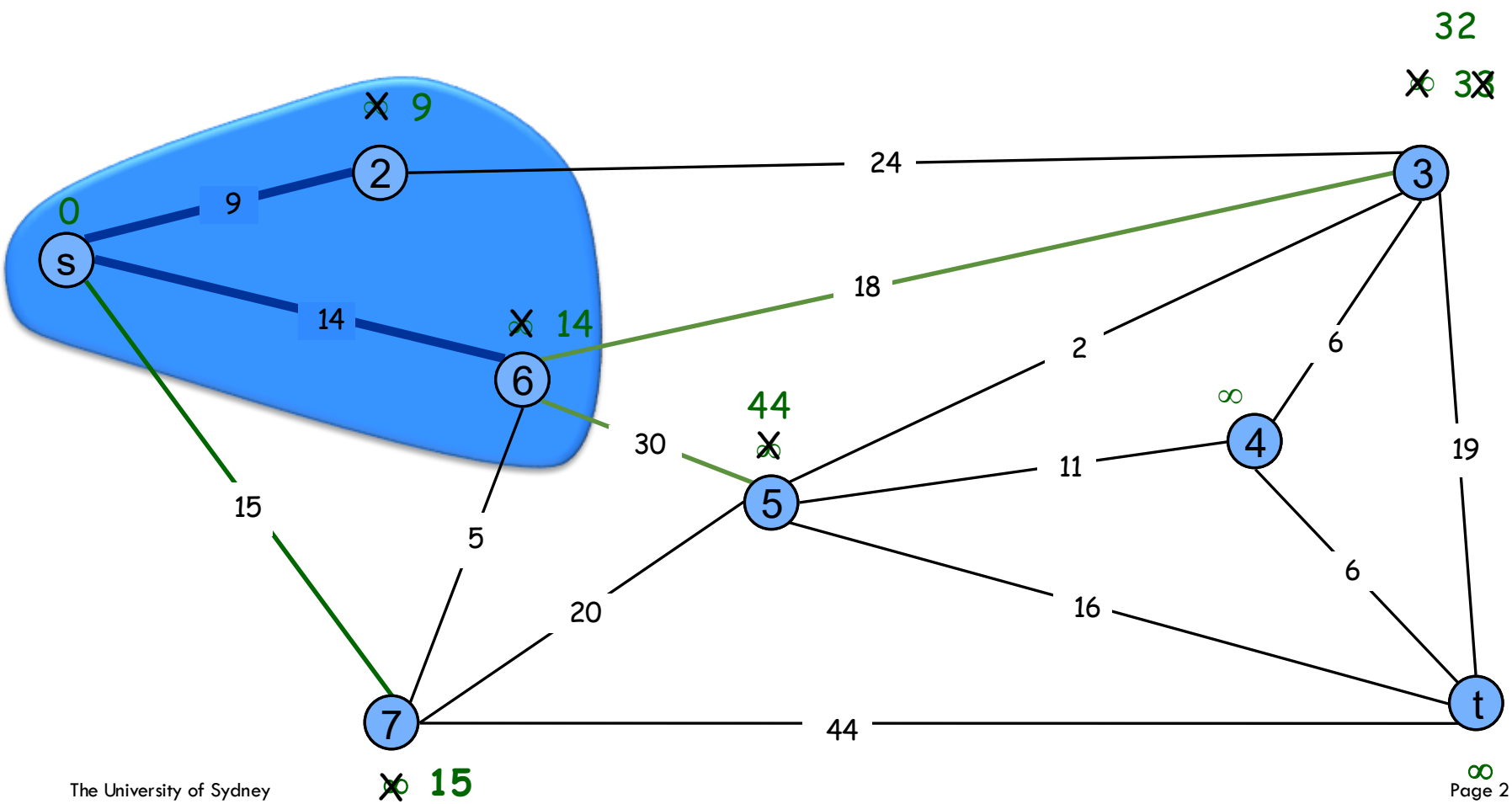


Dijkstra's Shortest Path Algorithm

S = { s, 2, 6 }
PQ = { 3, 4, 5, 7, t }



Dijkstra's Shortest Path Algorithm

S = { s, 2, 6 }
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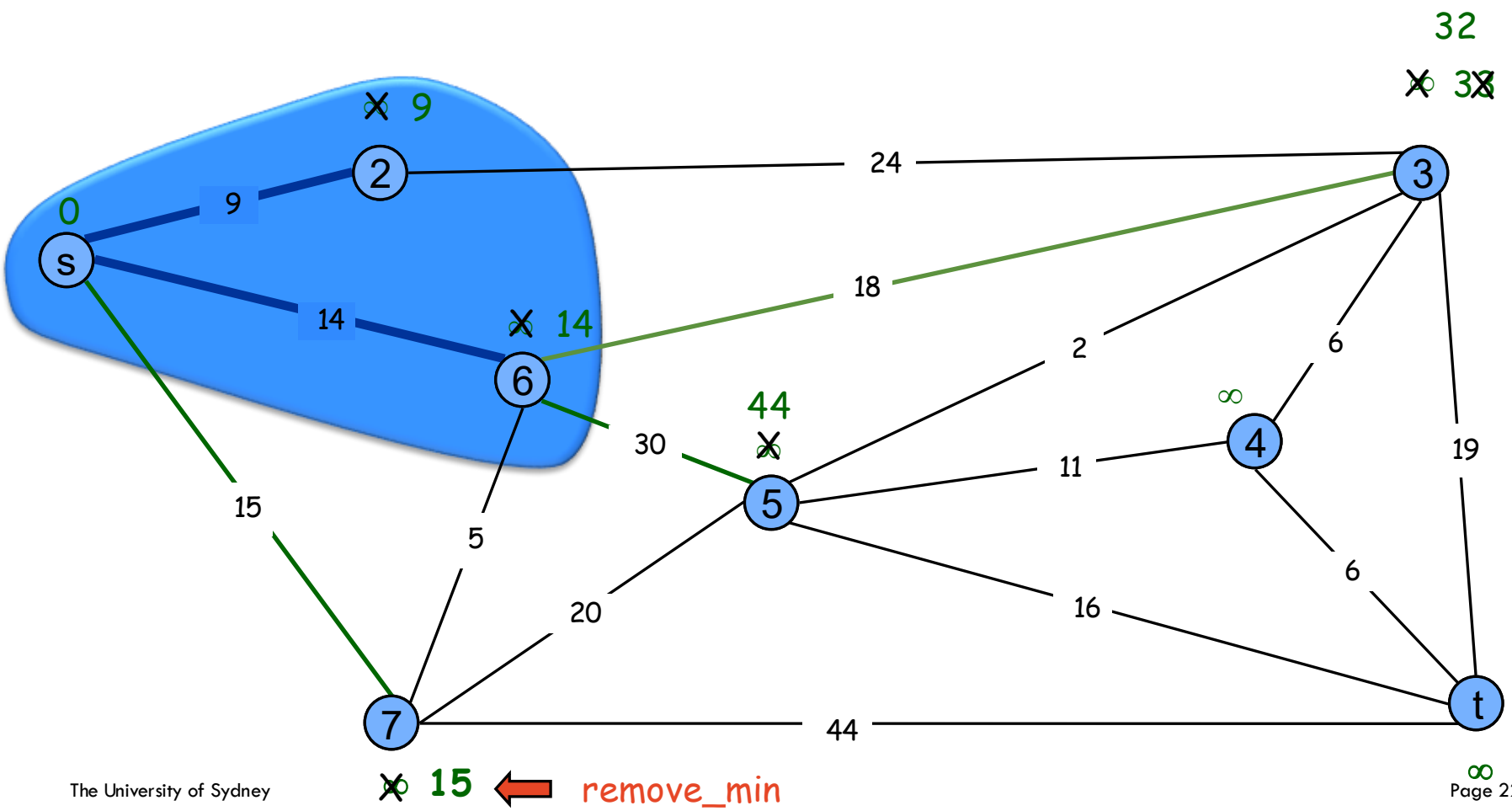

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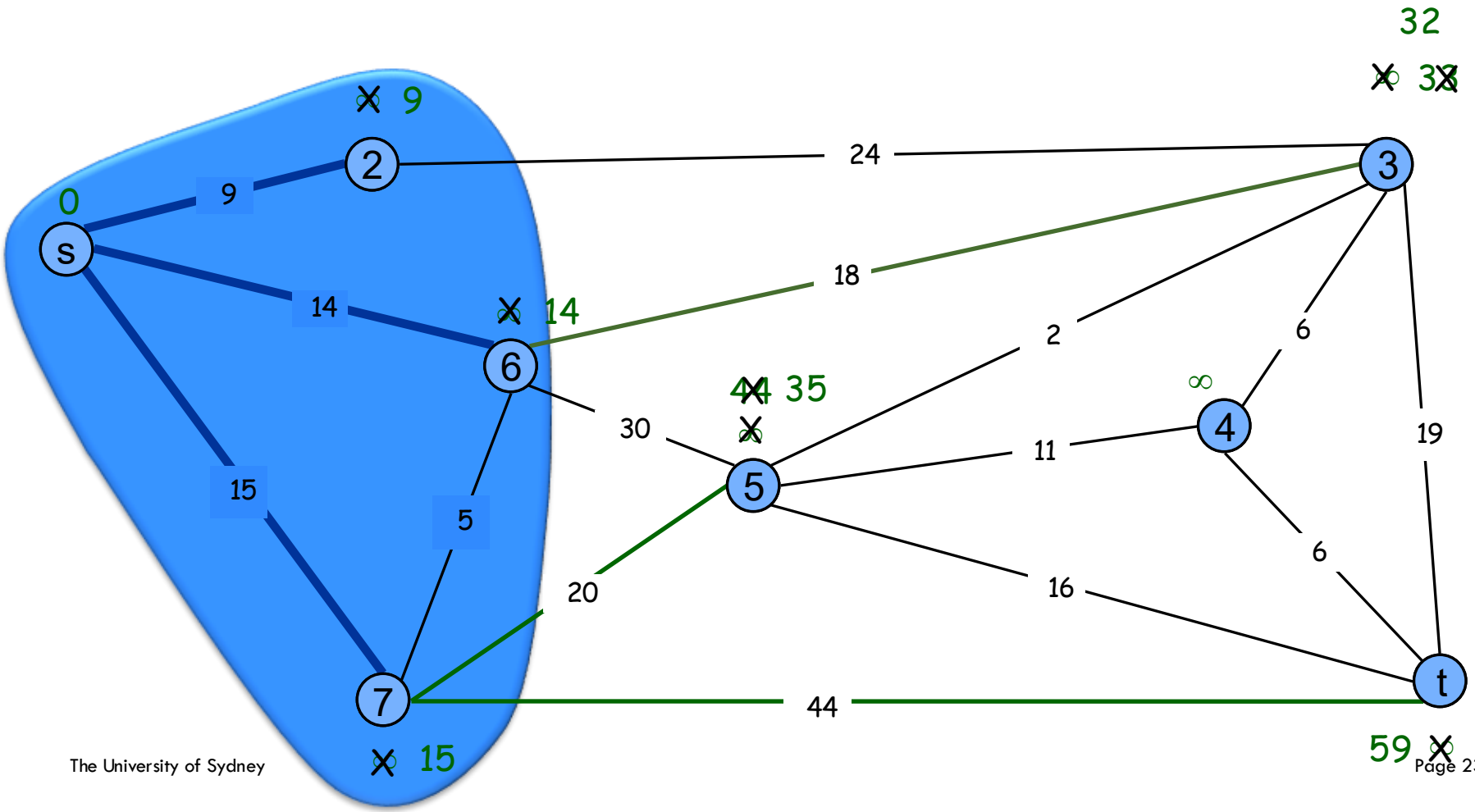

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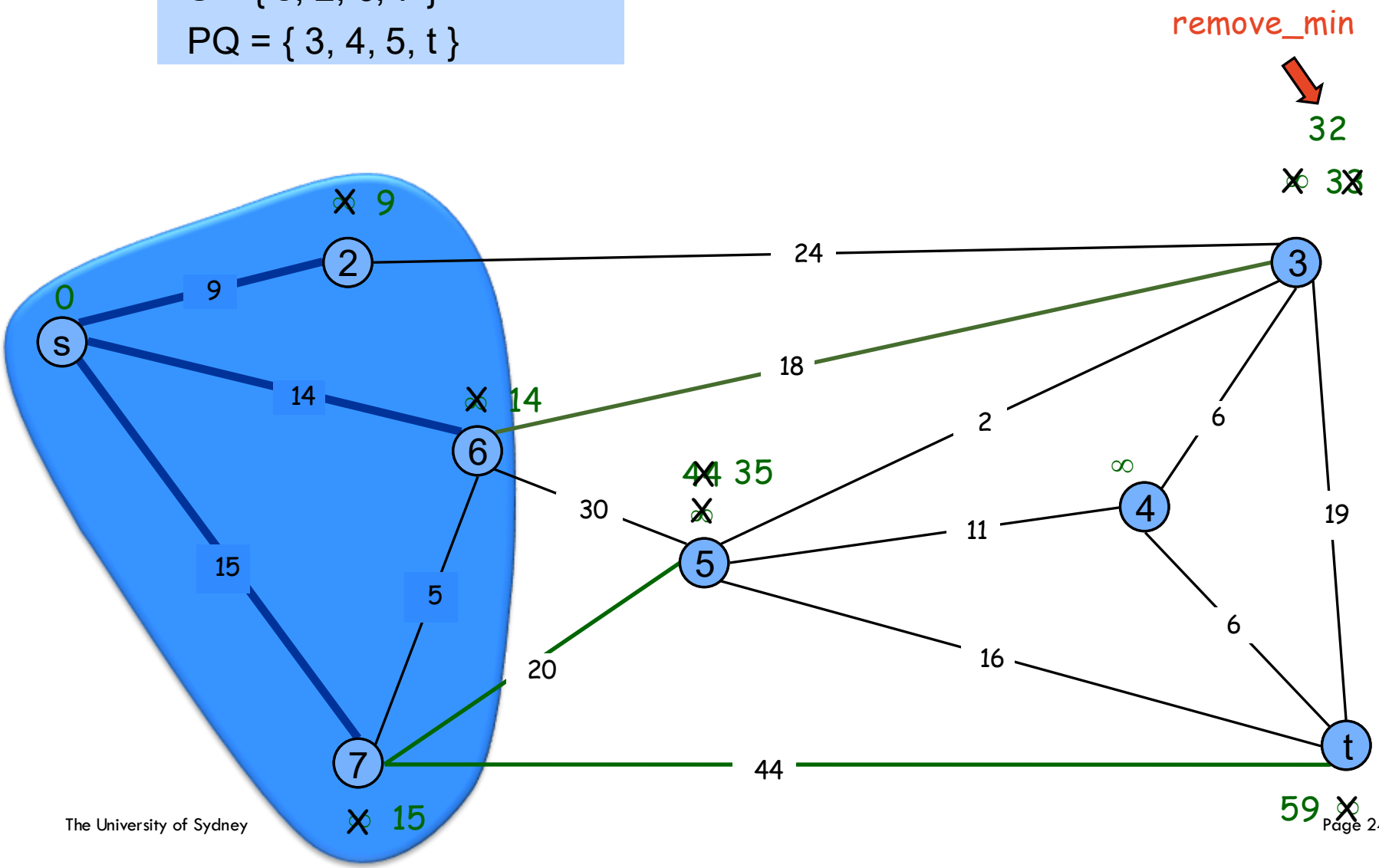

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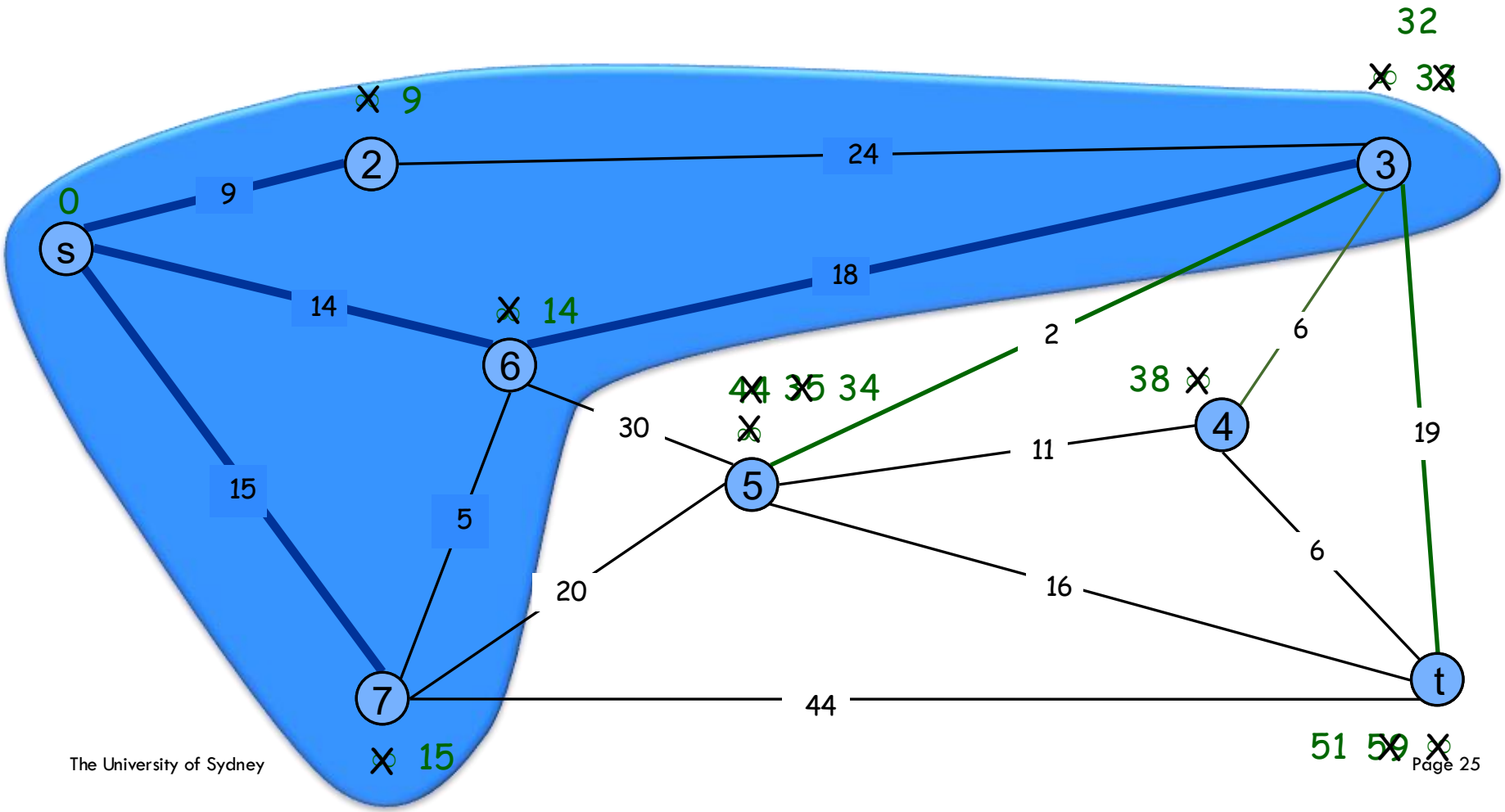

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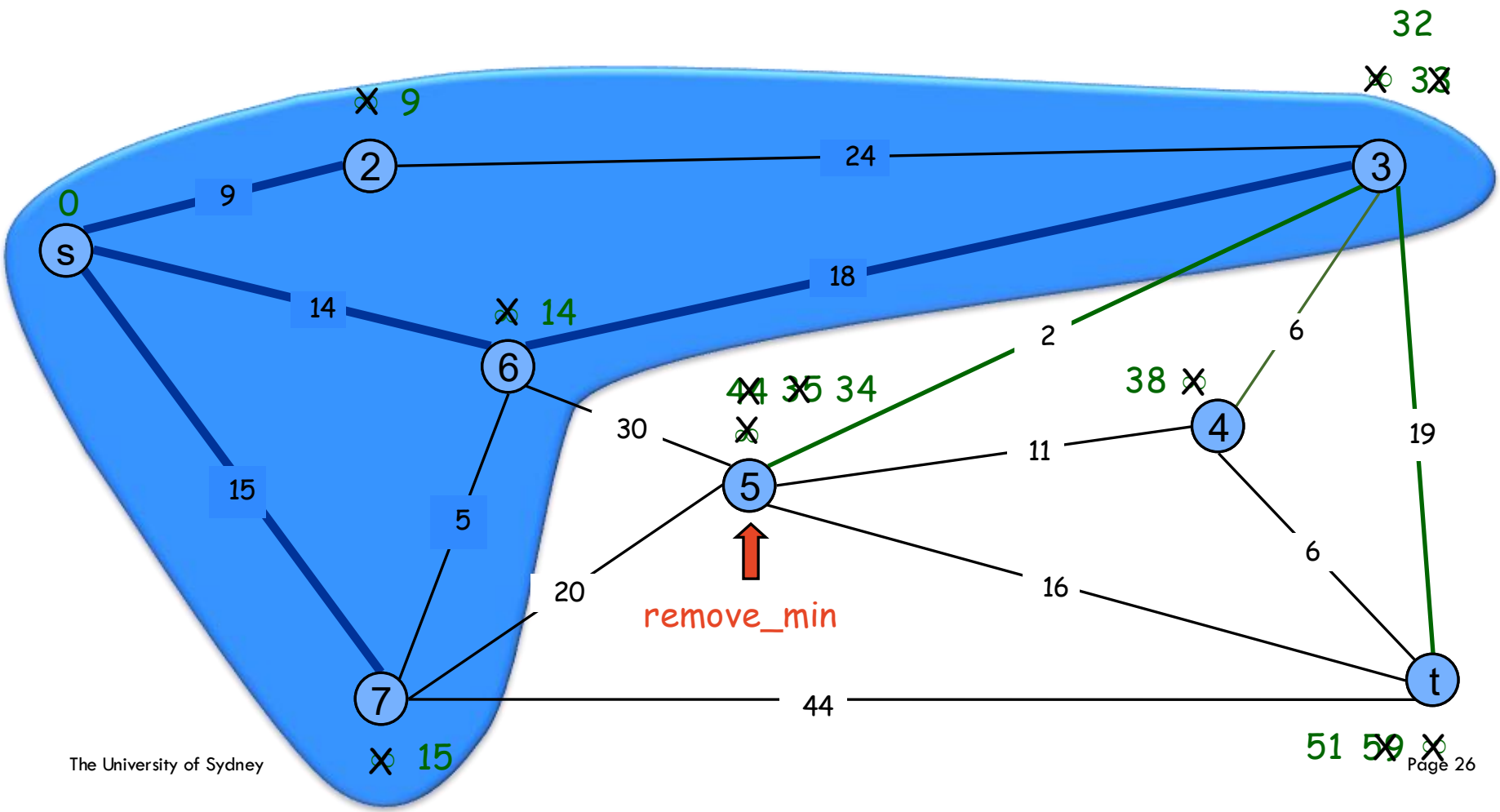

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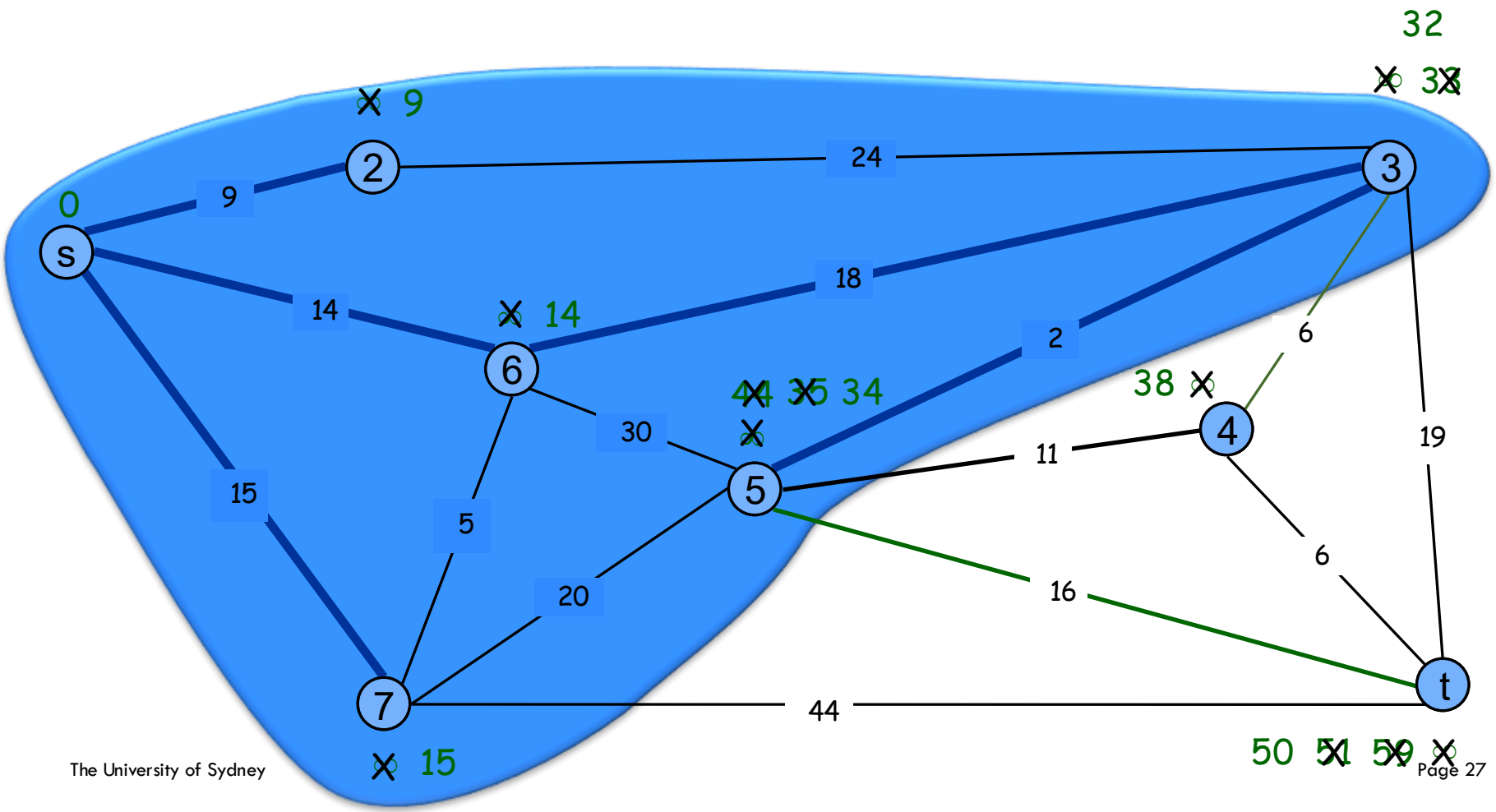

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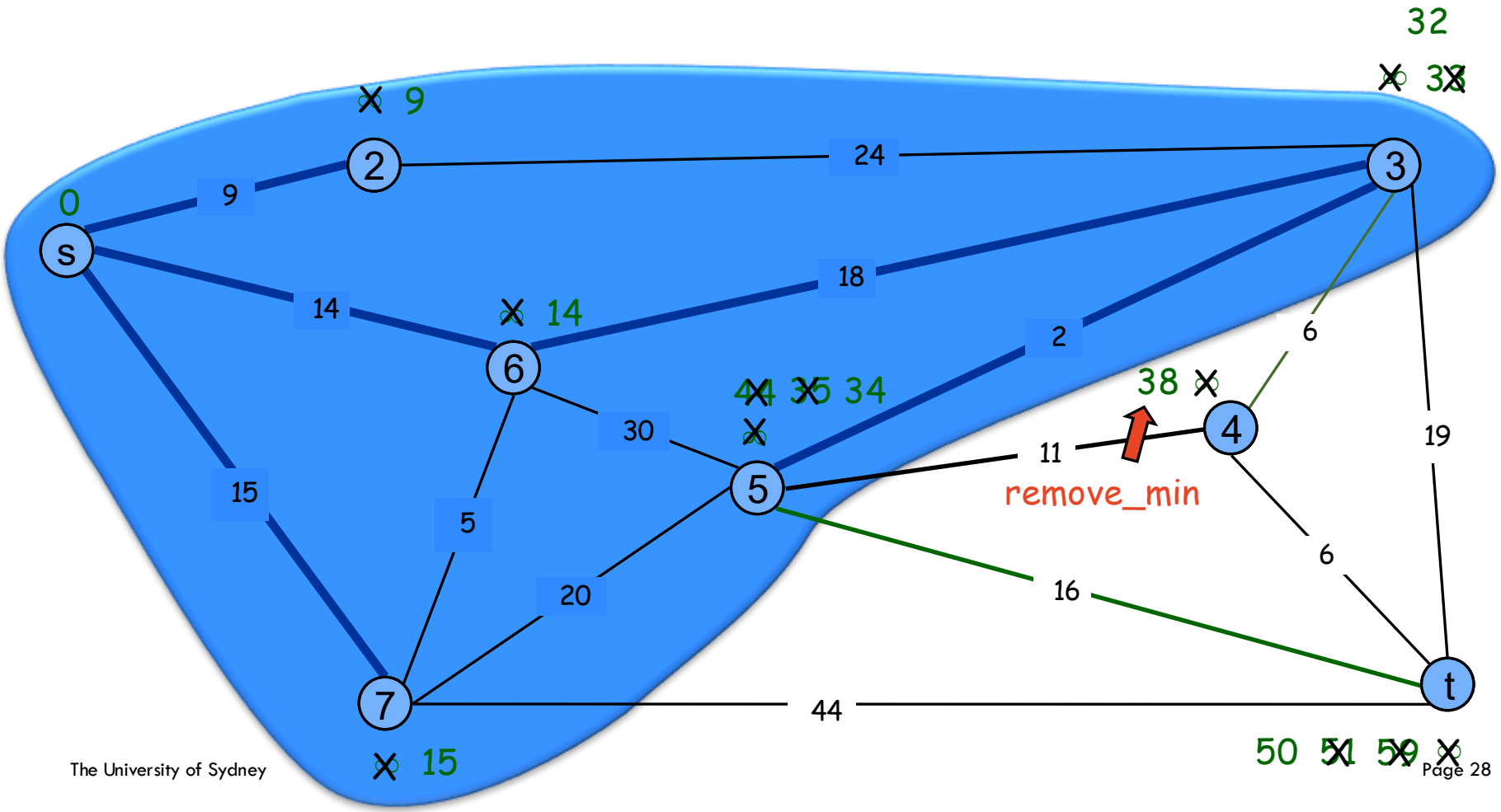

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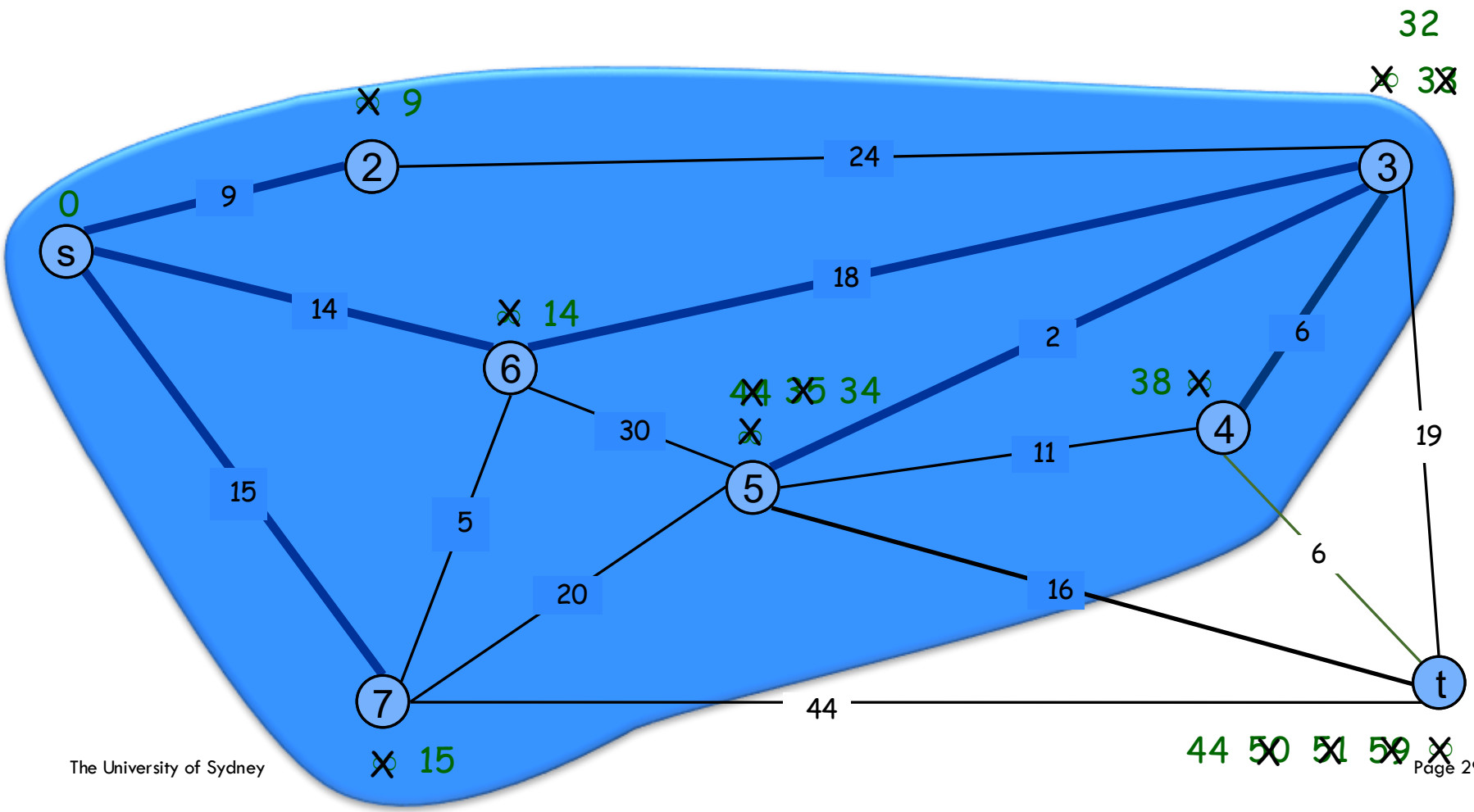

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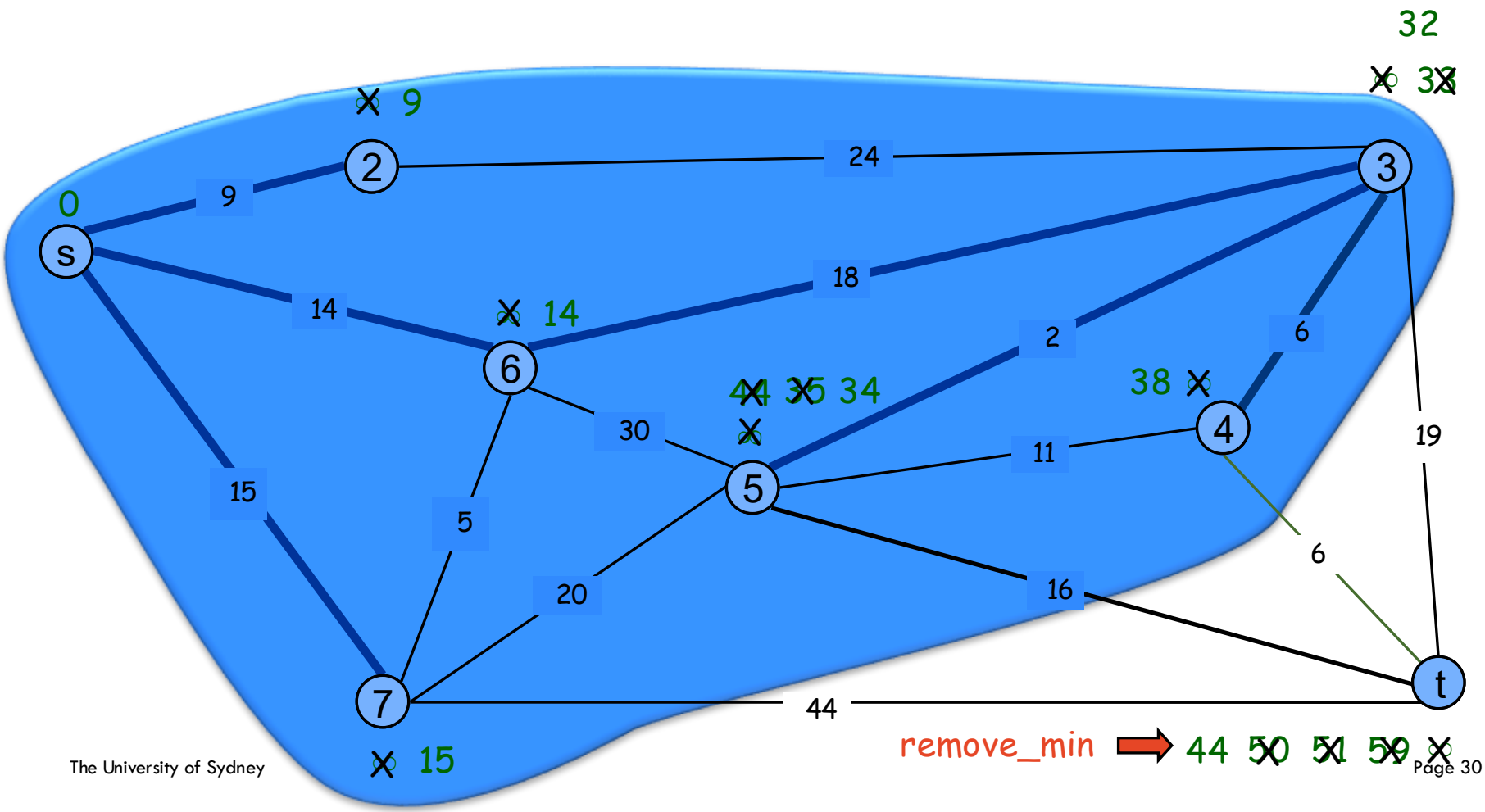

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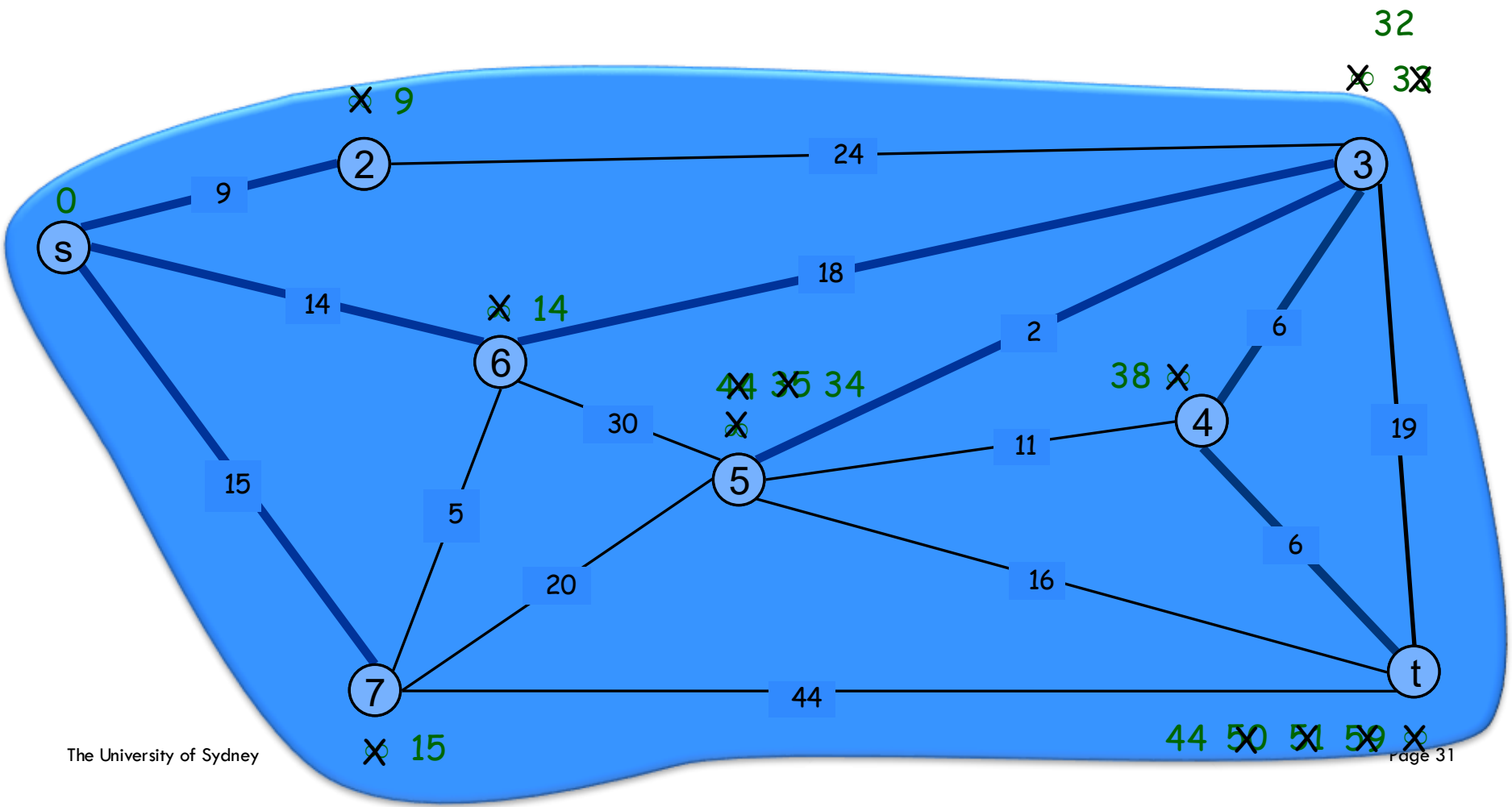

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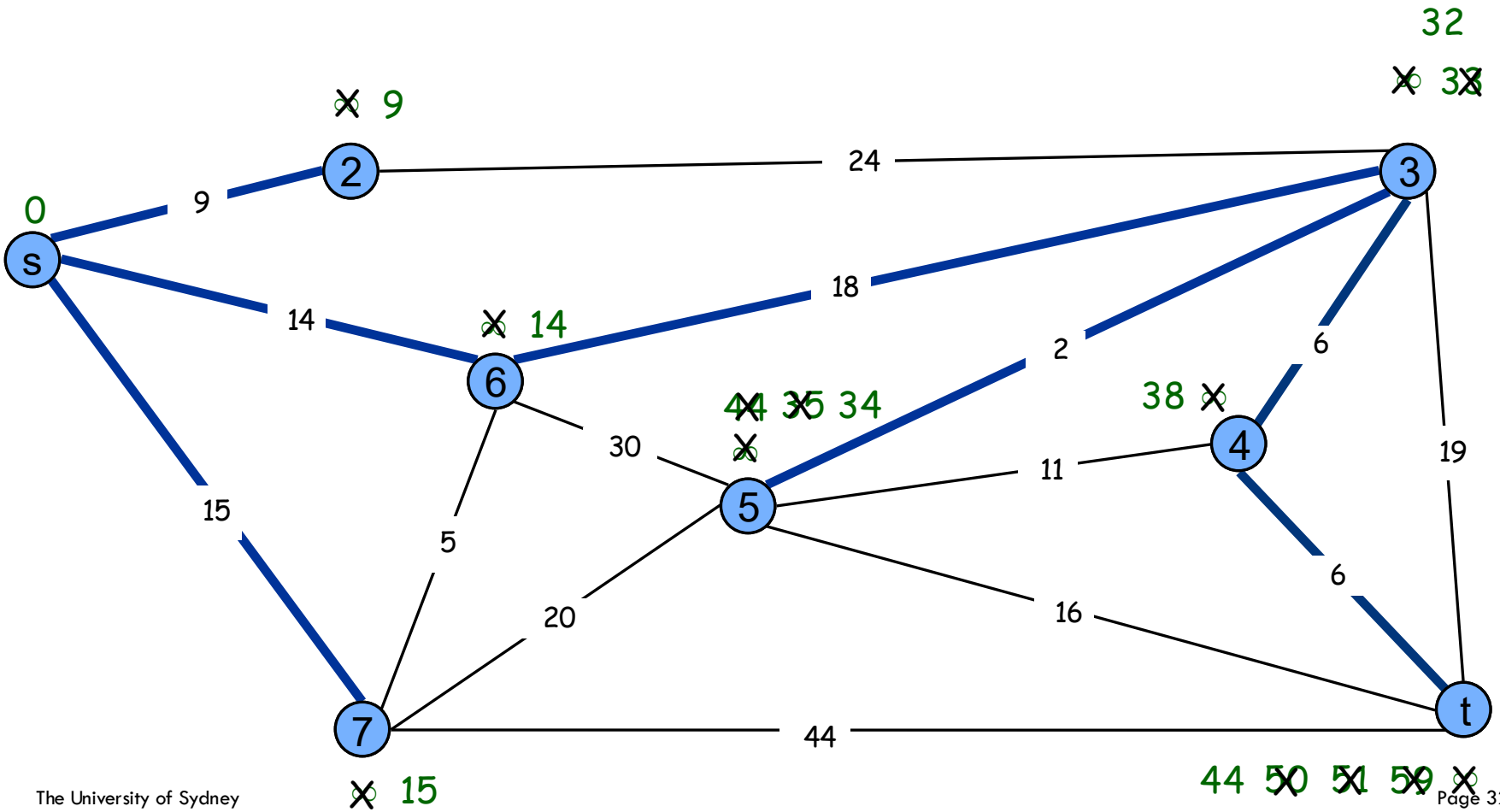
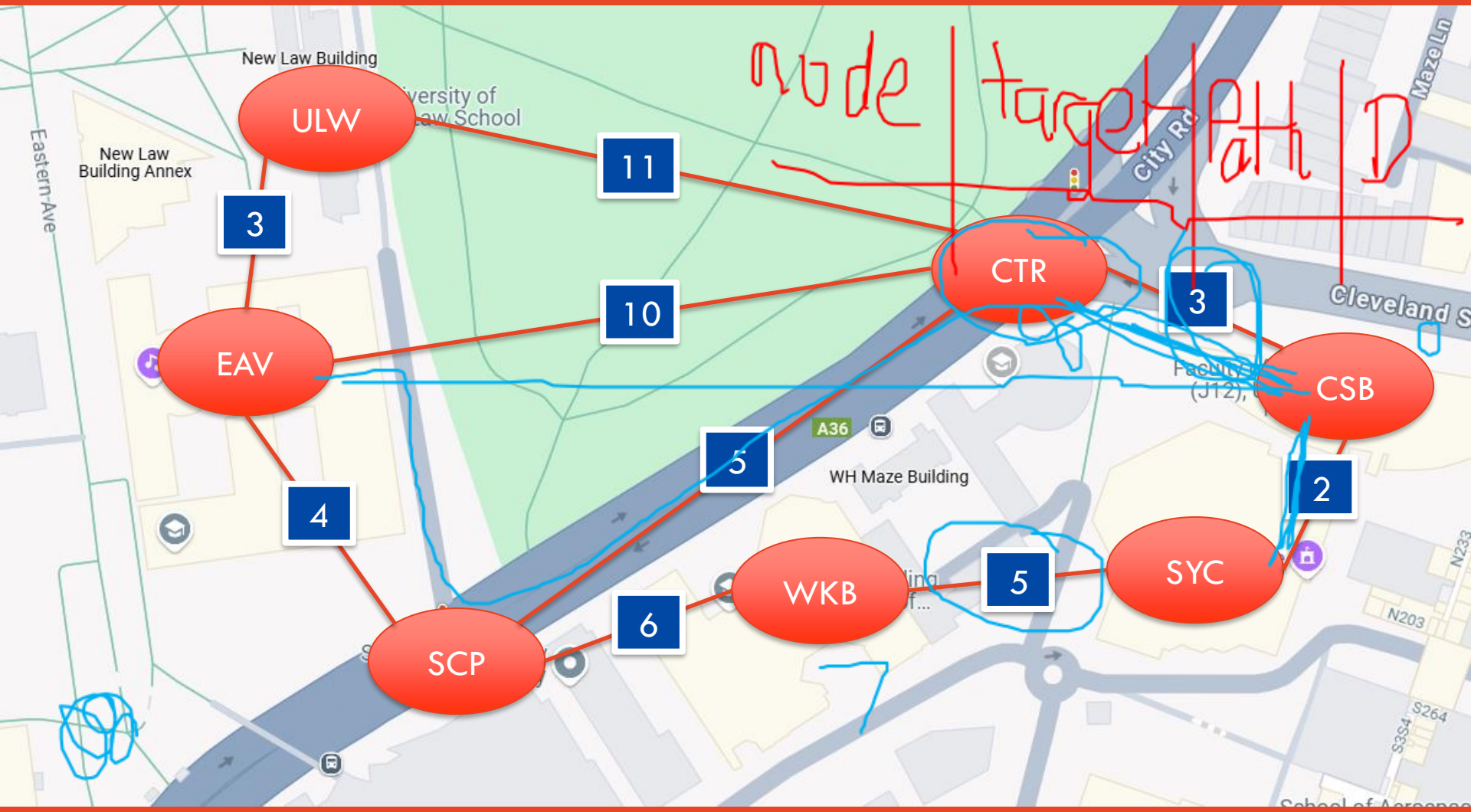


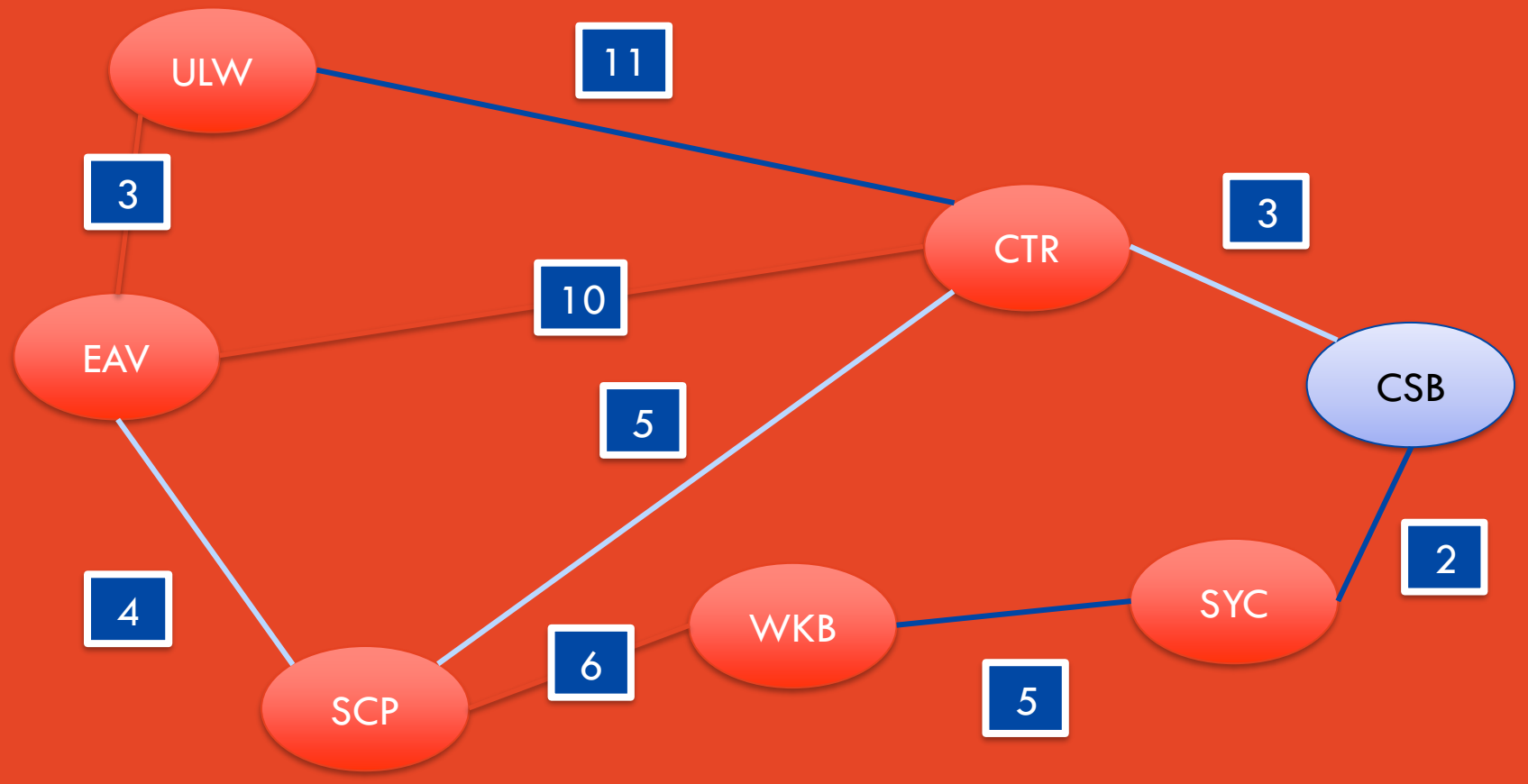
Challenge 9A: Use Djikstra to find the shortest path tree and then determine the shortest path from CSB to EAV



Challenge 9A: Use Djikstra to find the shortest path tree and then determine the shortest path from CSB to EAV

Node	Distance	Path
CSB	0	—
SYC	2	CSB → SYC
CTR	3	CSB → CTR
SCP	8	CSB → CTR → SCP
EAV	12	CSB → CTR → SCP → EAV
WKB	7	CSB → SYC → WKB
ULW	14	CSB → CTR → ULW

Challenge 9A: Use Djikstra to find the shortest path tree and then determine the shortest path from CSB to EAV



Dijkstra complexity analysis <u>except PQ ops</u>

```
def Dijkstra(G, w, s):

  # initialize algorithm
  for v in V do
    D[v] ← ∞
    parent[v] ← ∅
  D[s] ← 0
  Q ← new priority queue for { (v, D[v]) : v in V }

  # iteratively add vertices to S
  while Q is not empty do
    u ← Q.remove_min()
    for z in G.neighbors(u) do
      if D[u] + w[u, z] < D[z] then
        D[z] ← D[u] + w[u,z]
        Q.update_priority(z, D[z])
        parent[z] ← u
  return D, parent
```

O(n)

O(deg(u)) for each u in V
plus update_priority work

Dijkstra's Algorithm complexity analysis

Assuming the graph is connected (so $m \geq n-1$), the algorithm spends $O(m)$ time on everything except PQ operations

Priority queue operation counts:

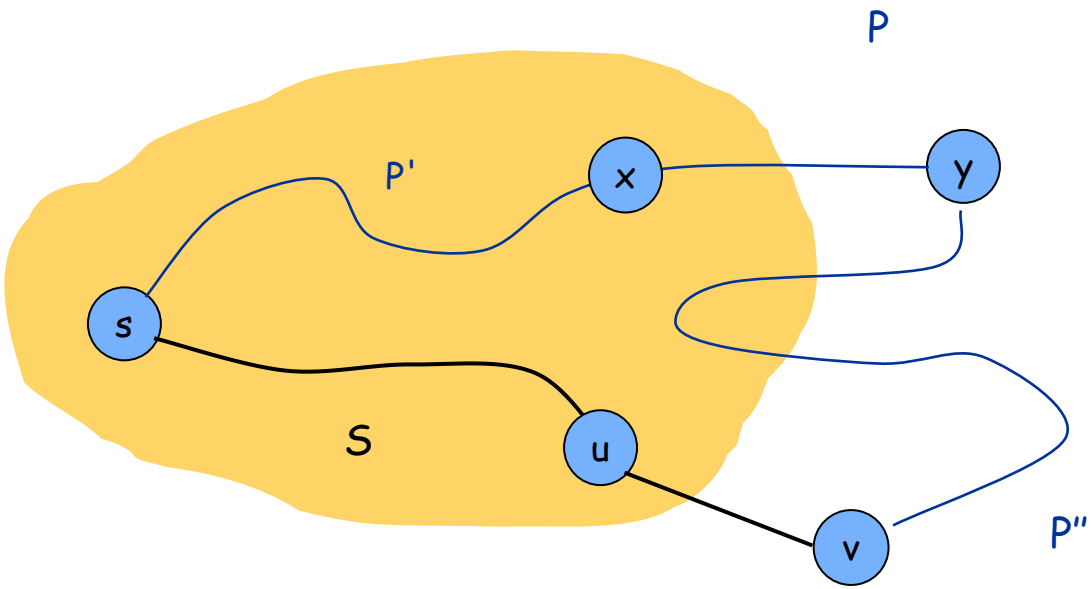
- insert: n
- decrease_key: m
- remove_min: n

Fact: Using a heap for PQ, Dijkstra runs in $O(m \log n)$ time

Fibonacci heap is a PQ that can carry out decrease key in $O(1)$ amortized time. Using that instead we get $O(m + n \log n)$ time.

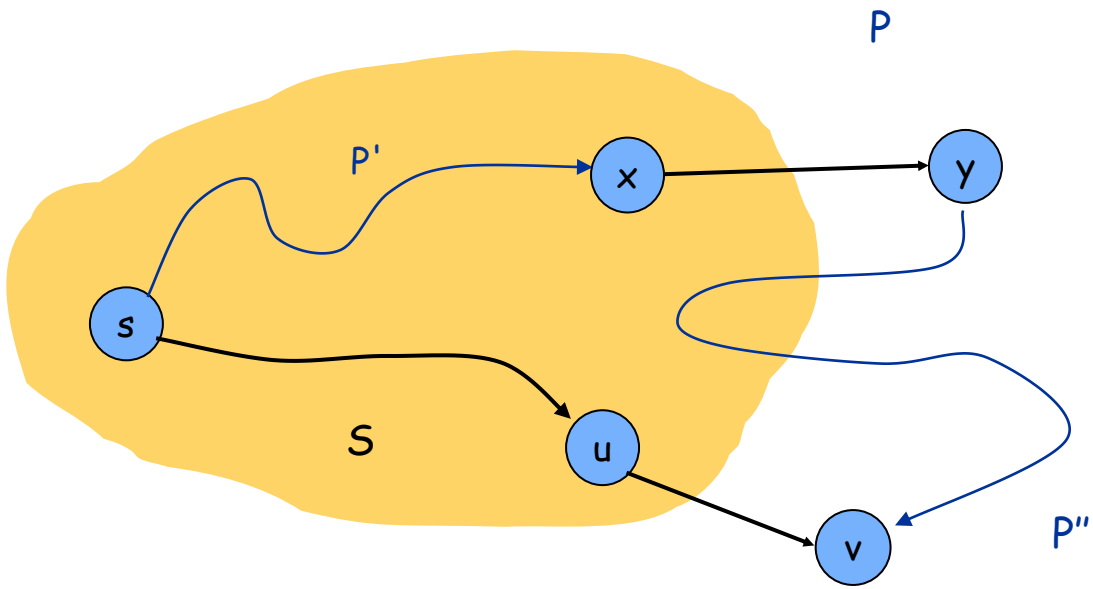
Warning: Dijkstra may not work for negative-weight edges

In the proof of correctness, even if D[v] is the smallest label, it may be that $dist_w(s, v) < D[v]$ if $w(P'') < 0$



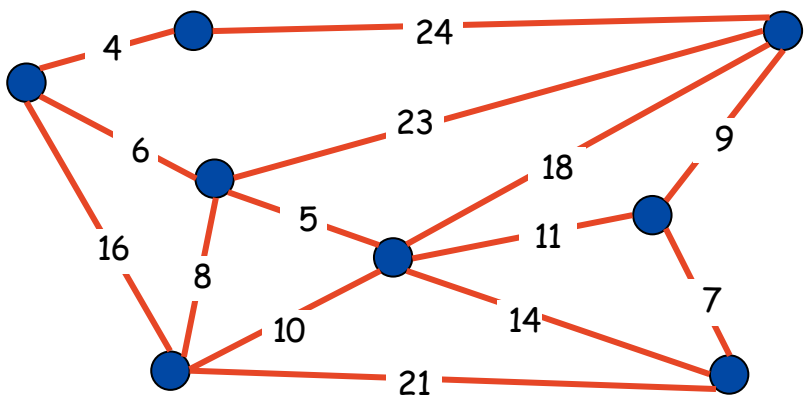
Dijkstra works for directed graph

In the proof of correctness, we need to use directed paths instead

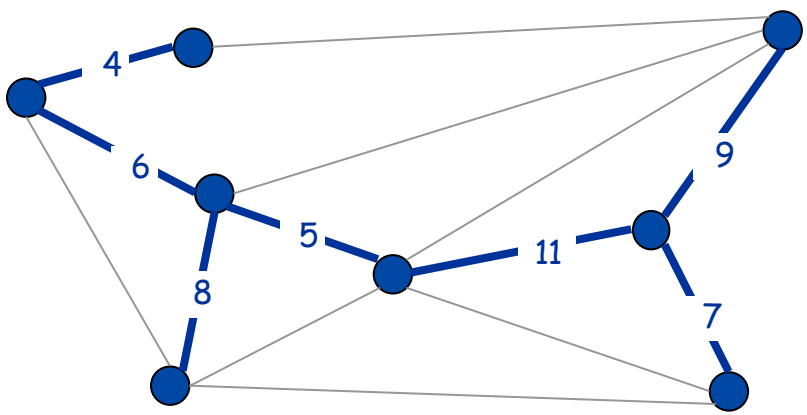


Minimum Spanning Tree

Given a connected graph G = (V, E) with real-valued edge weights c_e, an MST is a subset of the edges $T \subseteq E$ such that T is a spanning tree whose sum of edge weights is minimized.



G = (V, E)



T with $\sum_{e \in T} c_e = 50$

Applications

MST is fundamental problem with diverse applications.

Network design: Telephone, electrical, hydraulic, TV cable, computer, road

Approximation algorithms for NP-hard problems: traveling salesperson problem, Steiner tree

Indirect applications.

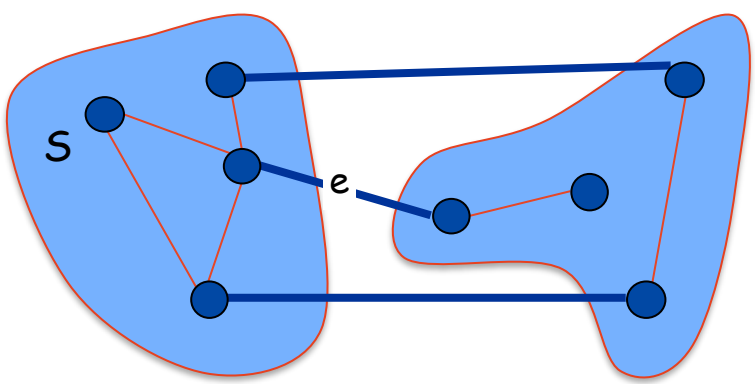
- max bottleneck paths
- LDPC codes for error correction
- image registration with Renyi entropy
- learning salient features for real-time face verification
- reducing data storage in sequencing amino acids in a protein
- …

MST properties

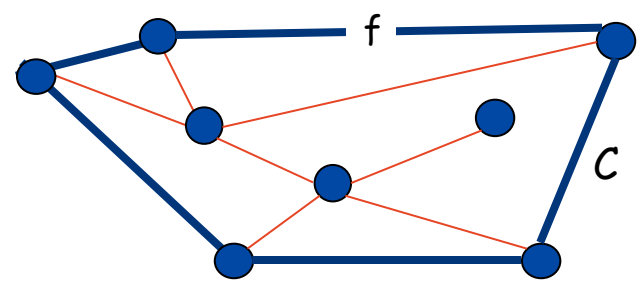
Simplifying assumption. All edge costs c_e are distinct.

Cut property. Let S be any subset of nodes, and let e be the min cost edge with exactly one endpoint in S. Then the MST contains e.

Cycle property. Let C be any cycle, and let f be the max cost edge belonging to C. Then the MST does not contain f.



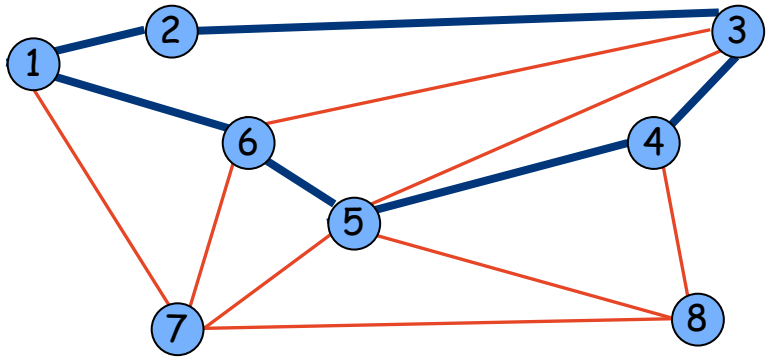
e is in the MST



f is not in the MST

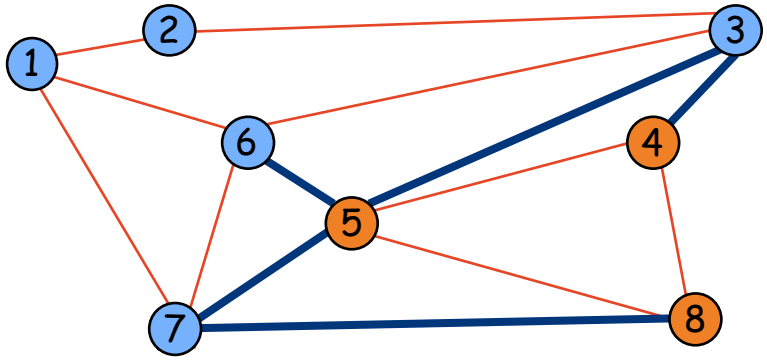
Cycles and Cuts

Cycle. Set of edges of the form a-b, b-c, c-d, …, y-z, z-a.



Cycle C = 1-2, 2-3, 3-4, 4-5, 5-6, 6-1

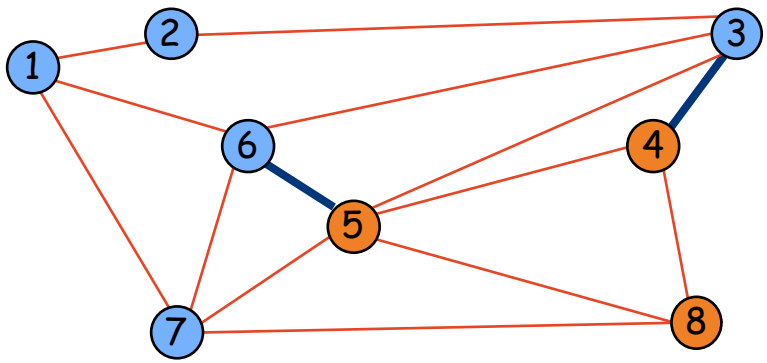
Cutset. A cut is a subset of nodes S. The corresponding cutset D is the subset of edges with exactly one endpoint in S.



Cut S = { 4, 5, 8 }
Cutset D = 5-6, 5-7, 3-4, 3-5, 7-8

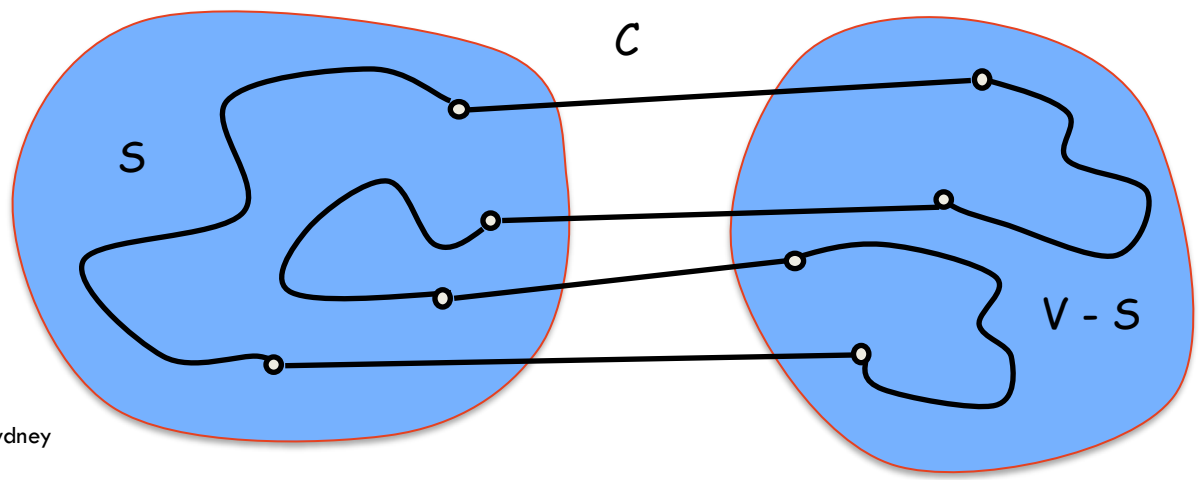
Cycle-Cut Intersection

Claim. A cycle and a cutset intersect in an even number of edges.



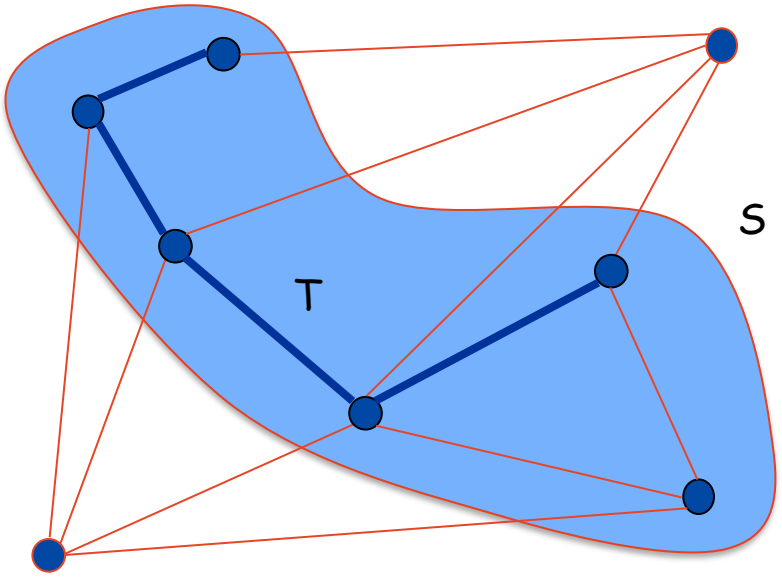
Cycle C = 1-2, 2-3, 3-4, 4-5, 5-6, 6-1
Cutset D = 3-4, 3-5, 5-6, 5-7, 7-8
Intersection = 3-4, 5-6

Proof:



Prim's Algorithm

```
def prim(G, c):
  u ← arbitrary vertex in V
  S ← { u }
  T ← Ø
  while |S| < |V| do
    (u, v) ← min cost edge s.t. u in S and v not in S
    add (u, v) to T
    add v to S
  return T
```



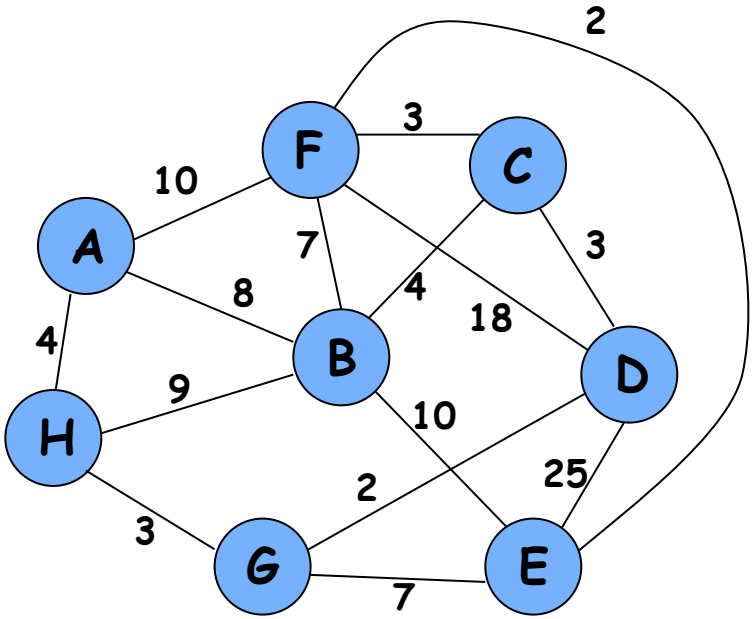
Implementation: Prim's Algorithm

```
def prim(G, c) {

 for v in V do
  d[v] ← ∞
  parent[v] ← ∅
 u ← arbitrary vertex in V
 d[u] ← 0
 Q ← new PQ with items { (v, d[v]) for v in V }
 S ← ∅

 while Q is not empty do
  u ← delete min element from Q
  add u to S
  for (u, v) incident to u do
   if v ∉ S and c_{u,v} < d[v] then
    parent[v] ← u
    decrease priority d[v] to c_{u,v}
return parent
```

Main idea: for every v in V \ S we keep

- d[v] = distance to closest neighbor in S
- parent[v] = closest neighbor in S

Walk-Through



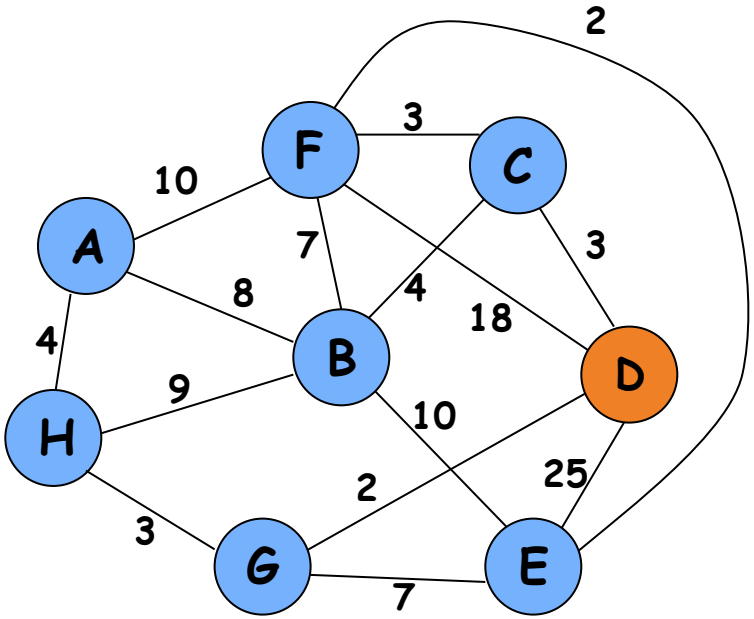
Initialize array

	S	d_v	p_v
A	F	∞	–
B	F	∞	–
C	F	∞	–
D	F	∞	–
E	F	∞	–
F	F	∞	–
G	F	∞	–
H	F	∞	–

Set S

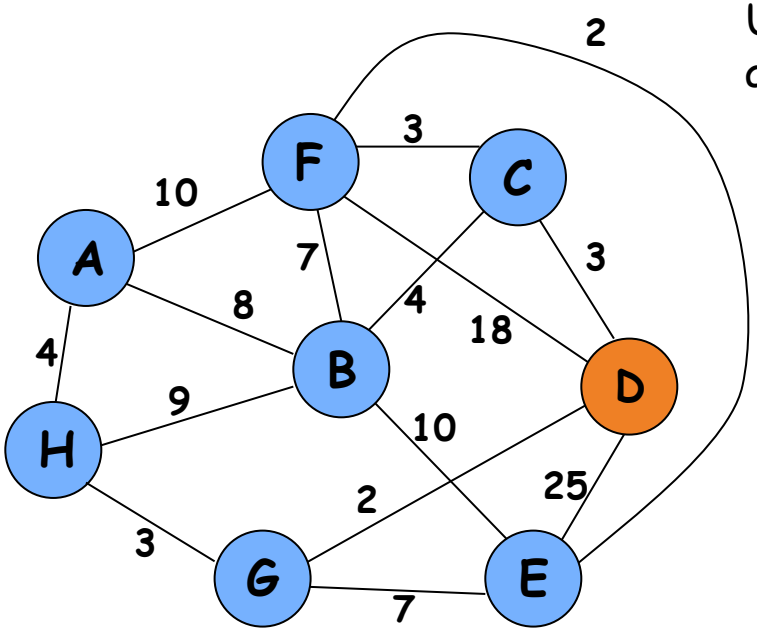
Min distance to S

Closest vertex in S



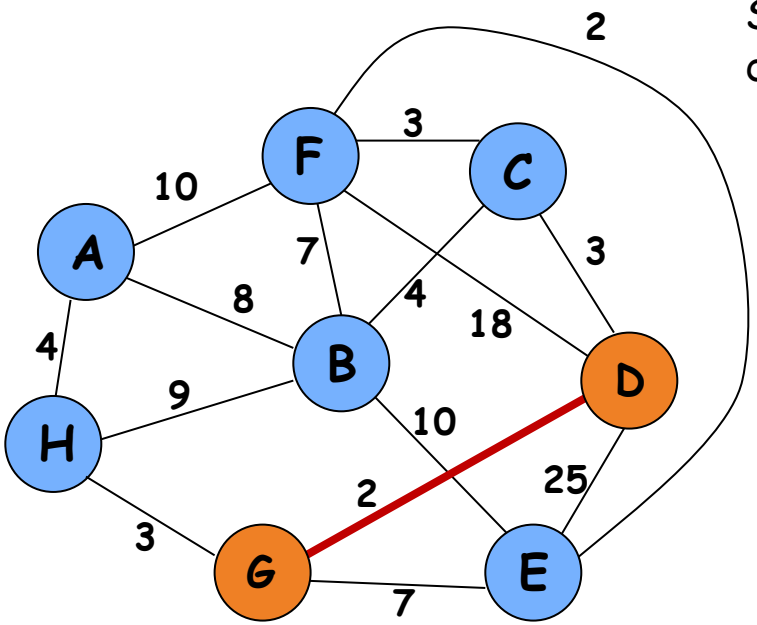
Start with any node, say D

	S	d_v	p_v
A			
B			
C			
D	T	0	–
E			
F			
G			
H			



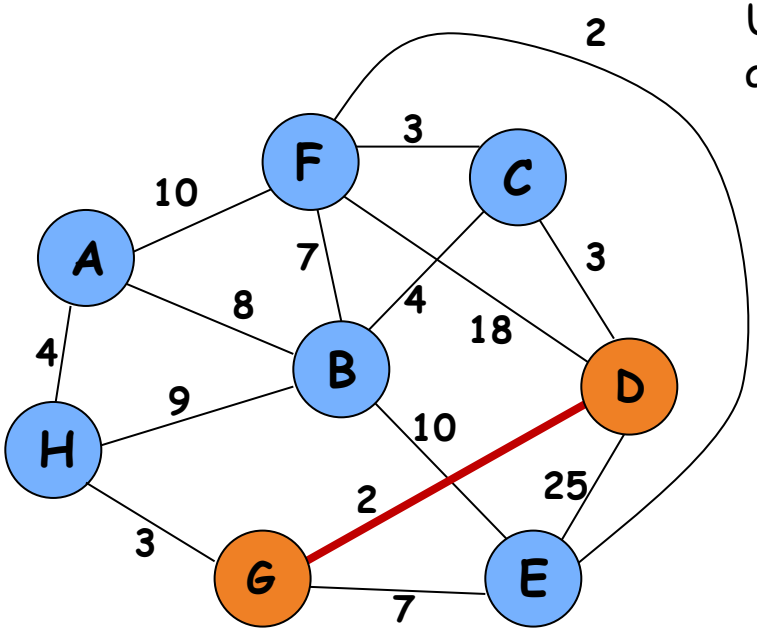
Update distances of adjacent, unselected nodes

	S	d_v	p_v
A			
B			
C		3	D
D	T	0	–
E		25	D
F		18	D
G		2	D
H			



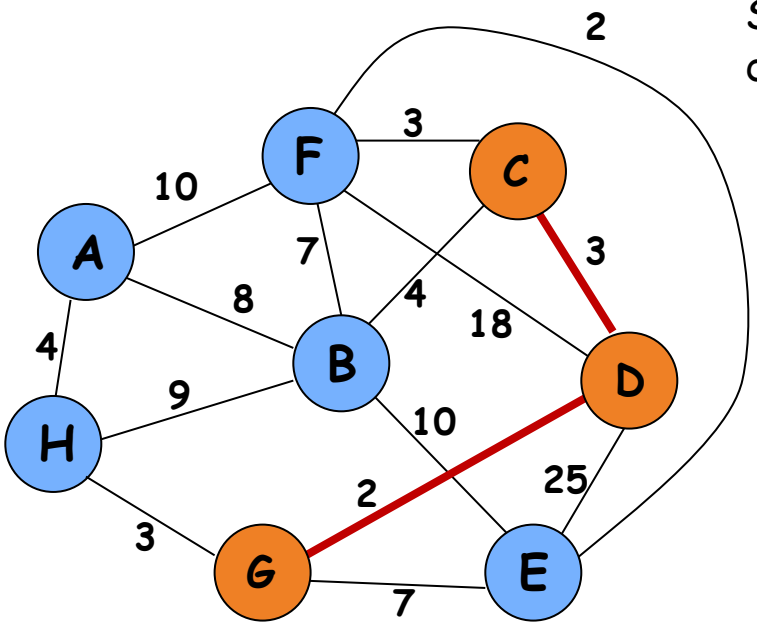
Select node with minimum distance

	S	d_v	p_v
A			
B			
C		3	D
D	T	0	–
E		25	D
F		18	D
G	T	2	D
H			



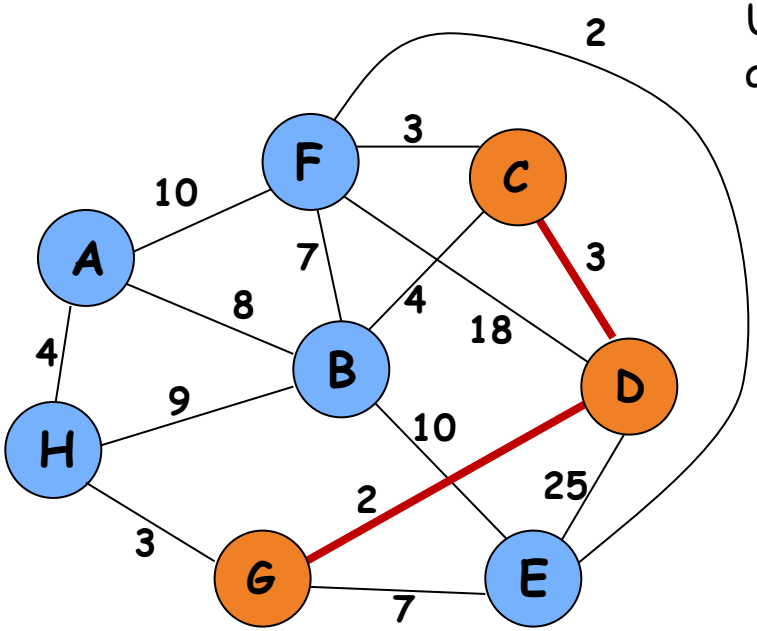
Update distances of adjacent, unselected nodes

	S	d_v	p_v
A			
B			
C		3	D
D	T	0	–
E		7	G
F		18	D
G	T	2	D
H		3	G



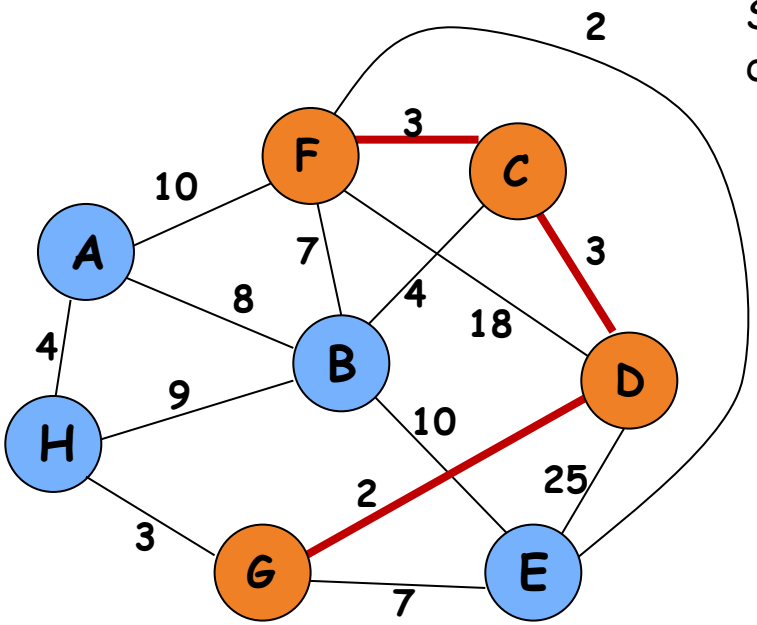
Select node with minimum distance

	S	d_v	p_v
A			
B			
C	T	3	D
D	T	0	–
E		7	G
F		18	D
G	T	2	D
H		3	G



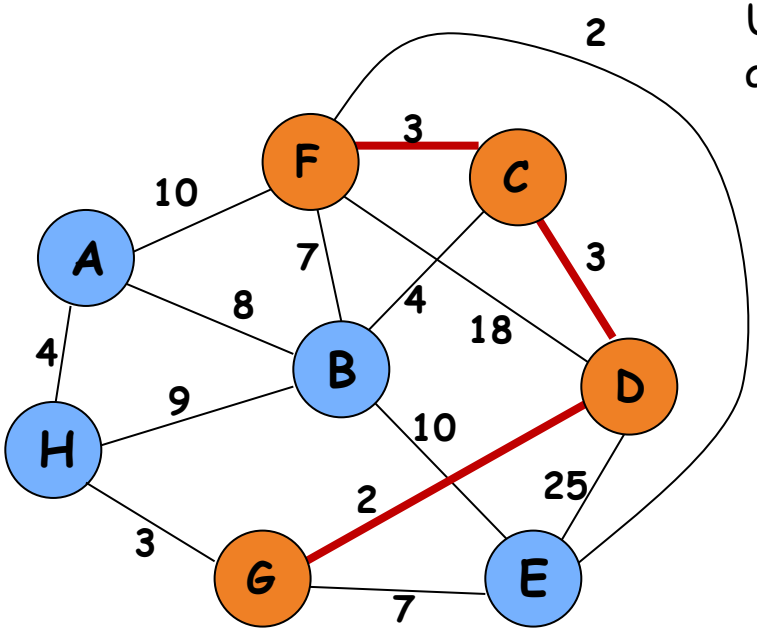
Update distances of adjacent, unselected nodes

	S	d_v	p_v
A			
B		4	C
C	T	3	D
D	T	0	–
E		7	G
F		3	C
G	T	2	D
H		3	G



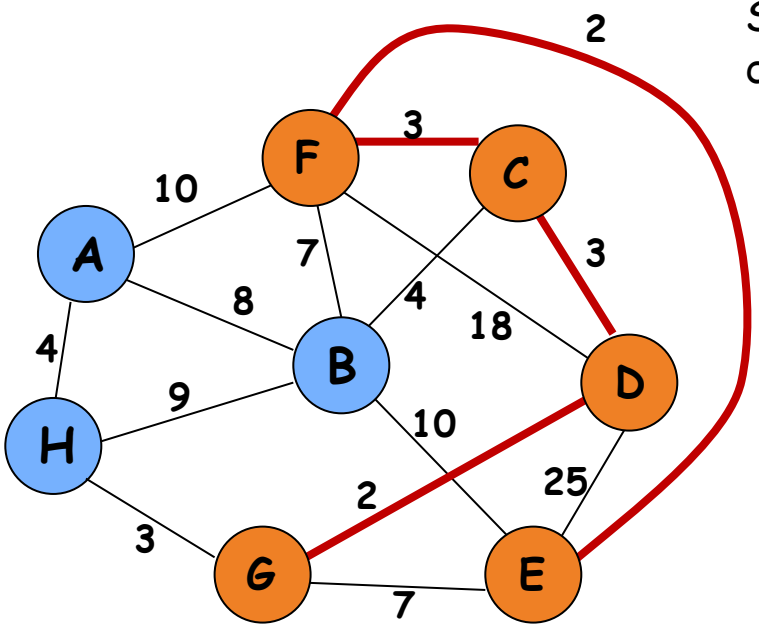
Select node with minimum distance

	S	d_v	p_v
A			
B		4	C
C	T	3	D
D	T	0	–
E		7	G
F	T	3	C
G	T	2	D
H		3	G



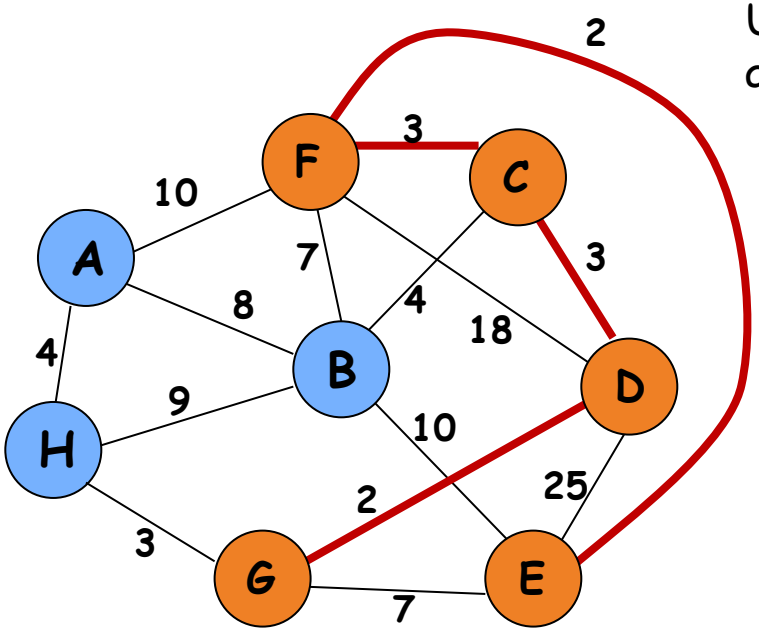
Update distances of adjacent, unselected nodes

	S	d_v	p_v
A		10	F
B		4	C
C	T	3	D
D	T	0	–
E		2	F
F	T	3	C
G	T	2	D
H		3	G



Select node with minimum distance

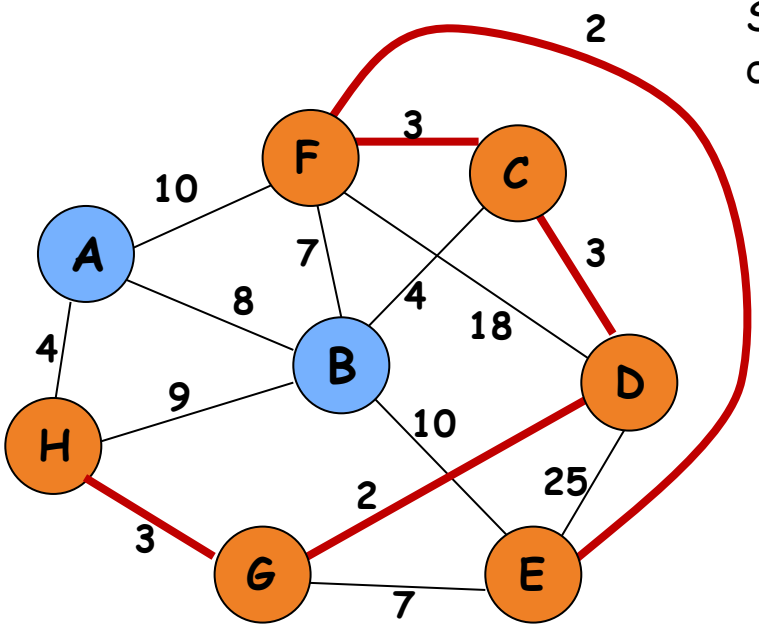
	S	d_v	p_v
A		10	F
B		4	C
C	T	3	D
D	T	0	–
E	T	2	F
F	T	3	C
G	T	2	D
H		3	G



Update distances of adjacent, unselected nodes

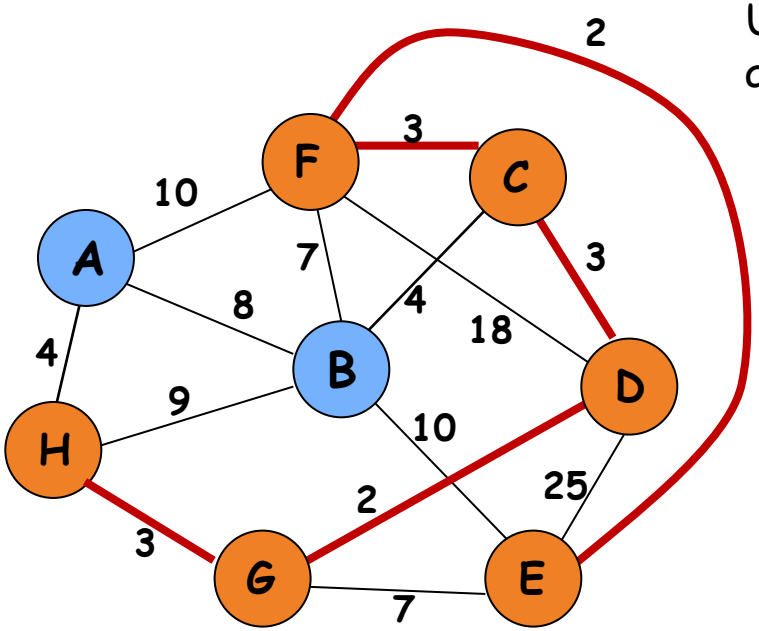
	S	d_v	p_v
A		10	F
B		4	C
C	T	3	D
D	T	0	–
E	T	2	F
F	T	3	C
G	T	2	D
H		3	G

Table entries unchanged



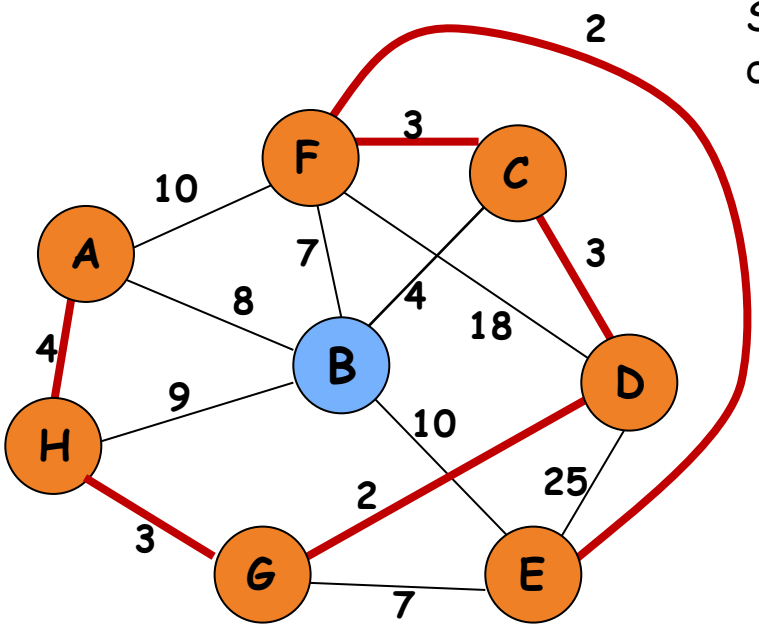
Select node with minimum distance

	S	d_v	p_v
A		10	F
B		4	C
C	T	3	D
D	T	0	–
E	T	2	F
F	T	3	C
G	T	2	D
H	T	3	G



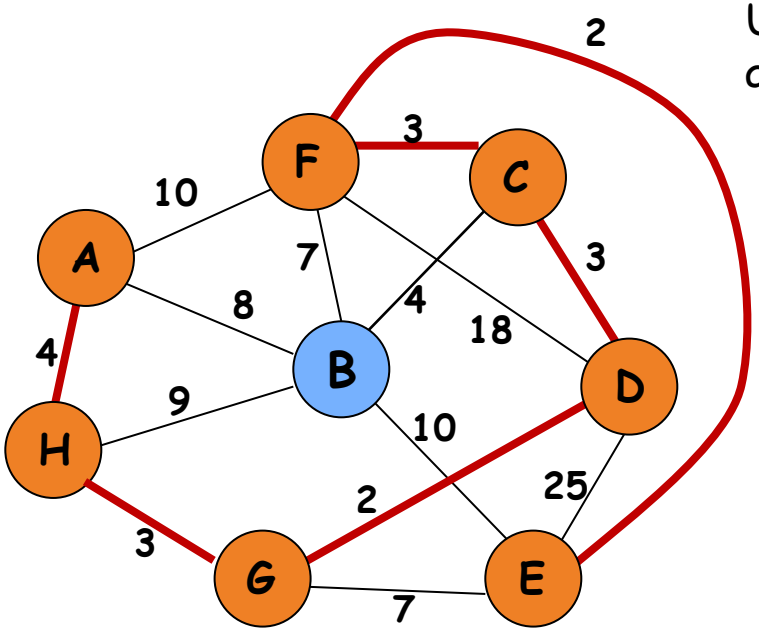
Update distances of adjacent, unselected nodes

	S	d_v	p_v
A		4	H
B		4	C
C	T	3	D
D	T	0	–
E	T	2	F
F	T	3	C
G	T	2	D
H	T	3	G



Select node with minimum distance

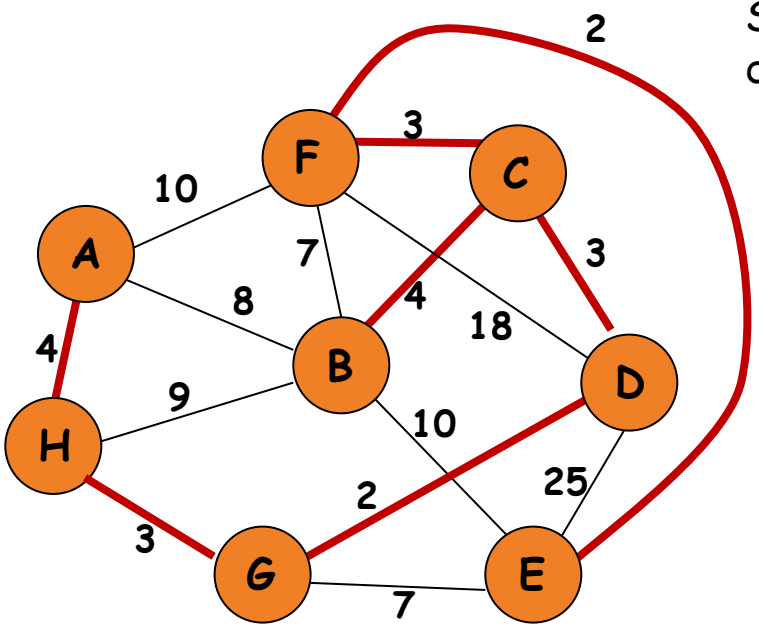
	S	d_v	p_v
A	T	4	H
B		4	C
C	T	3	D
D	T	0	–
E	T	2	F
F	T	3	C
G	T	2	D
H	T	3	G



Update distances of adjacent, unselected nodes

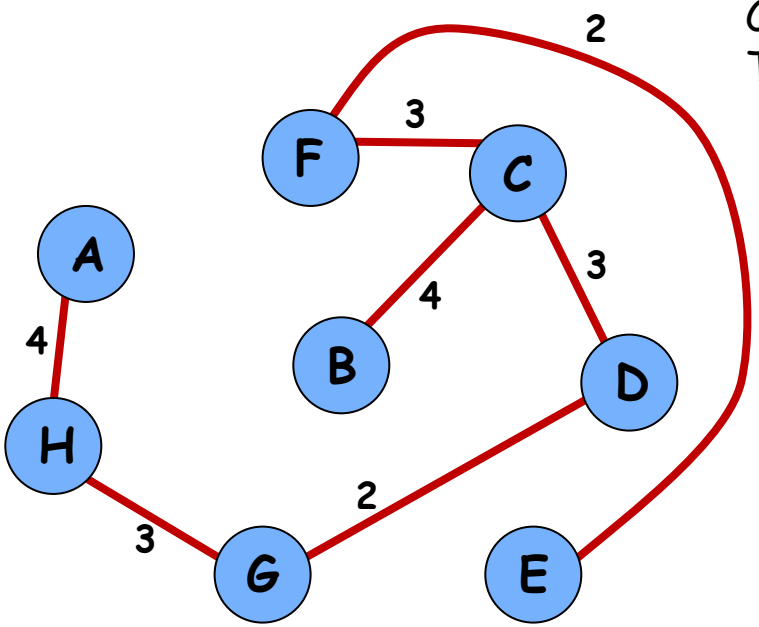
	S	d_v	p_v
A	T	4	H
B		4	C
C	T	3	D
D	T	0	–
E	T	2	F
F	T	3	C
G	T	2	D
H	T	3	G

Table entries unchanged



Select node with minimum distance

	S	d_v	p_v
A	T	4	H
B	T	4	C
C	T	3	D
D	T	0	–
E	T	2	F
F	T	3	C
G	T	2	D
H	T	3	G



Cost of Minimum Spanning Tree = $\Sigma\, d_v$ = 21

	S	d_v	p_v
A	T	4	H
B	T	4	C
C	T	3	D
D	T	0	–
E	T	2	F
F	T	3	C
G	T	2	D
H	T	3	G

Done!

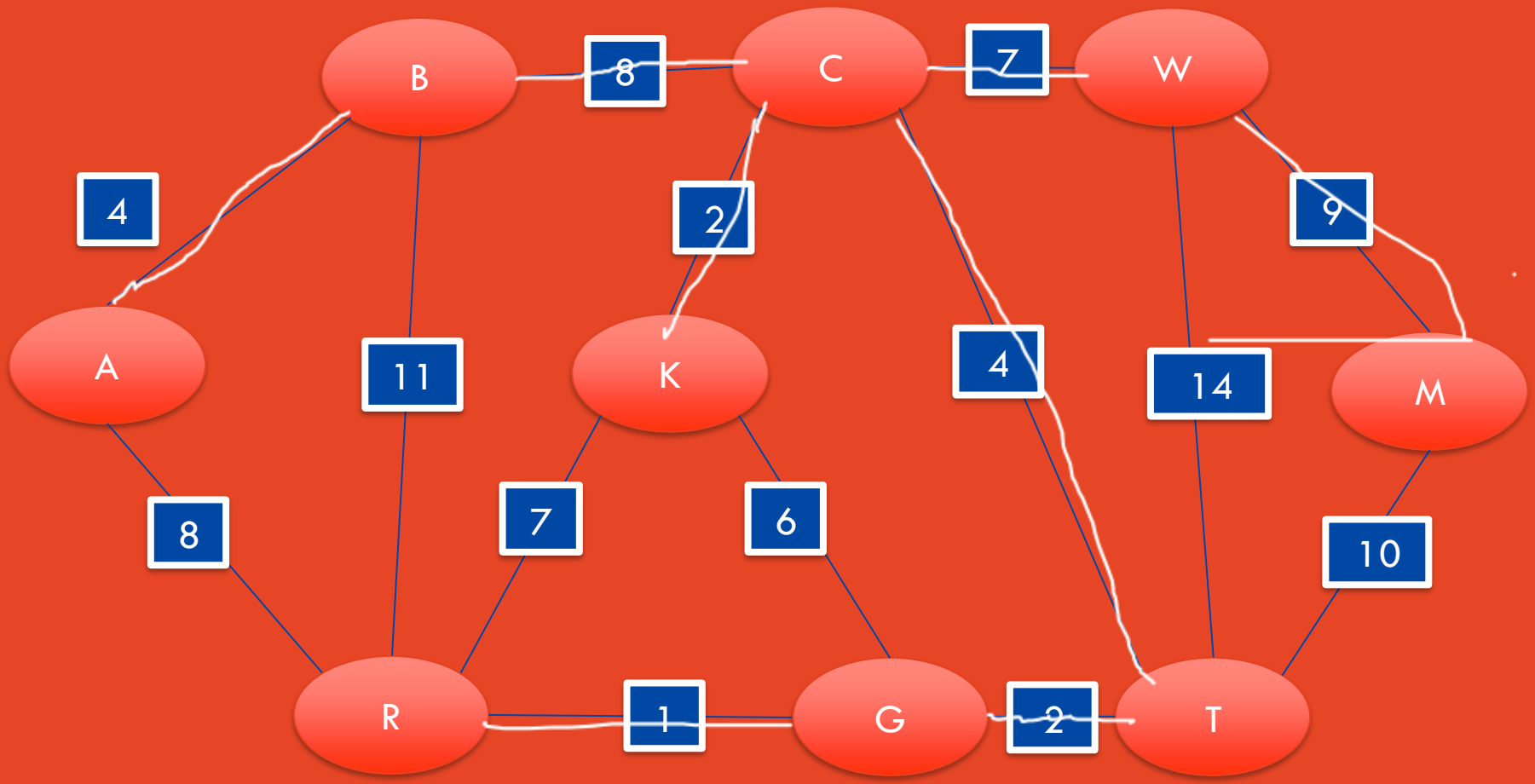
Prim's Algorithm complexity

```
def prim(G, c) {

 for v in V do
  d[v] ← ∞
  parent[v] ← ∅
 u ← arbitrary vertex in V
 d[u] ← 0
 Q ← new PQ with items { (v, d[v]) for v in V }
 S ← ∅

 while Q is not empty do
  u ← delete min element from Q
  S ← S ∪ { u }
  for (u, v) incident to u do
   if v ∉ S and c_{u,v} < d[v] then
    parent[v] ← u
    decrease priority d[v] to c_{u,v}
 return parent
```

Similar analysis to Dijkstra's algorithm:

- O(m log n) using a heap
- O(m + n log n) using Fibonacci heap

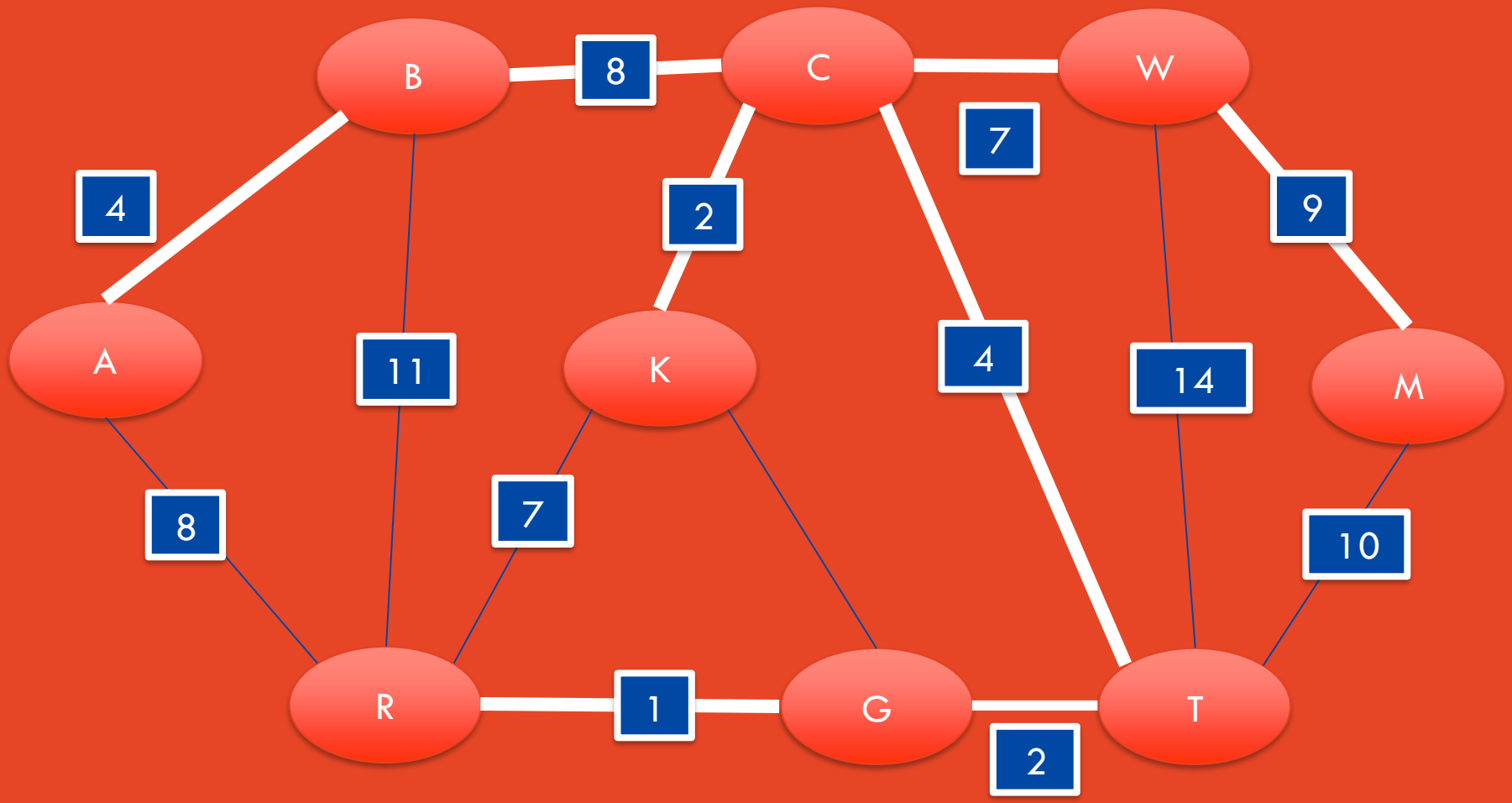
Challenge 9B: Prims MST



You are building an eco-friendly village – in order to do this you must use Prim's Algorithm to find the MST for this Graph

Inspired by: EducateYourself Channel

Challenge 9B: Prims MST



You are building an eco-friendly village – in order to do this you must use Prim's Algorithm to find the MST for this Graph

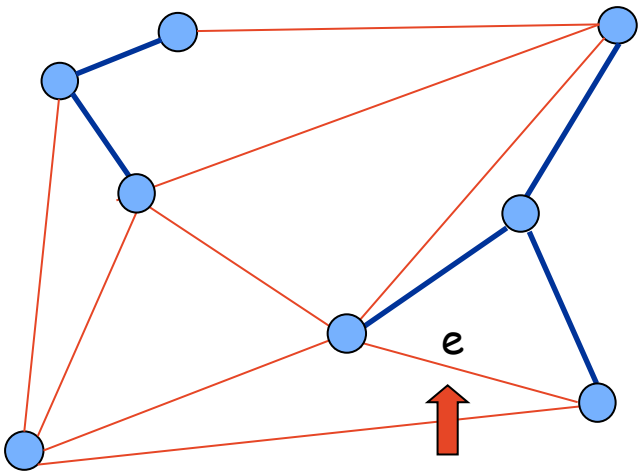
Inspired by: EducateYourself Channel

Kruskal's Algorithm

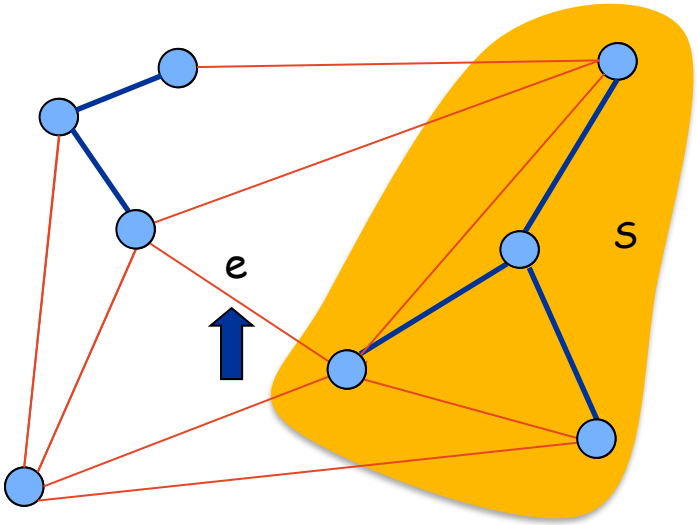
Consider edges in ascending order of weight.

Case 1: If adding e to T creates a cycle, discard e according to cycle property.

Case 2: Otherwise, insert e = (u, v) into T according to cut property where S = set of nodes in u's connected component.

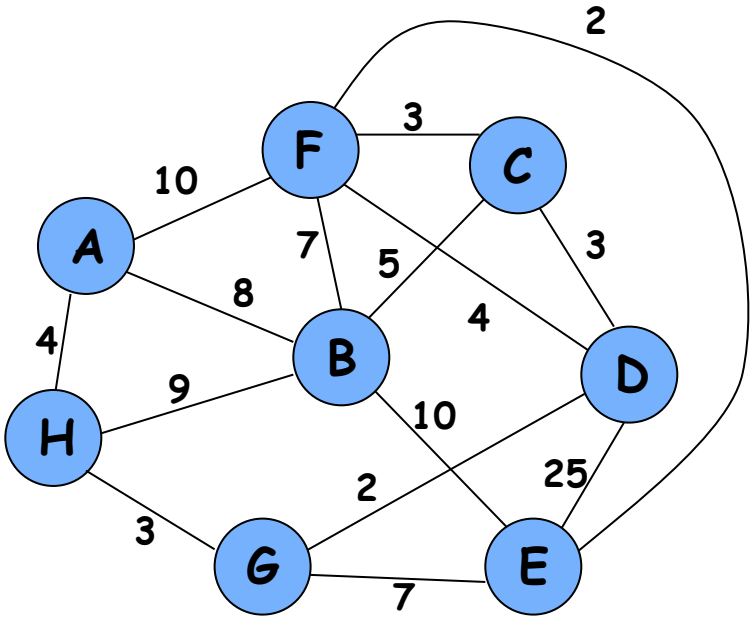


Case 1

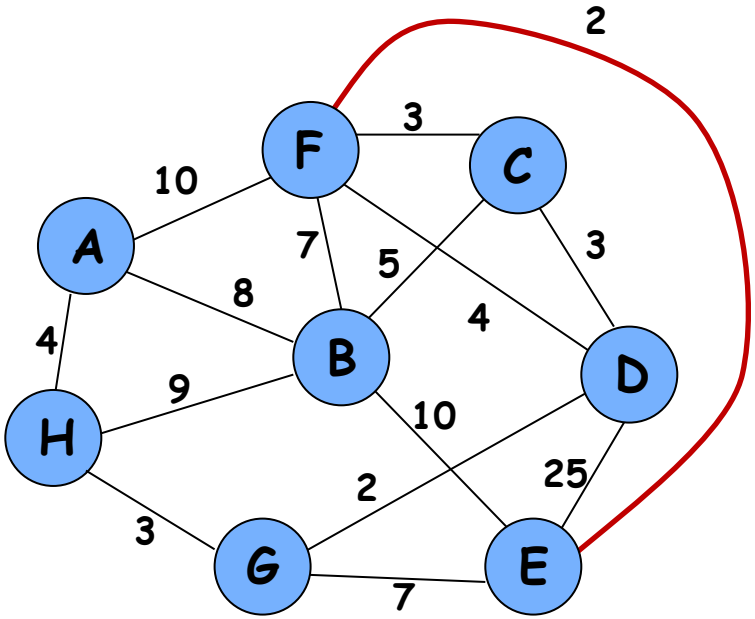


Case 2

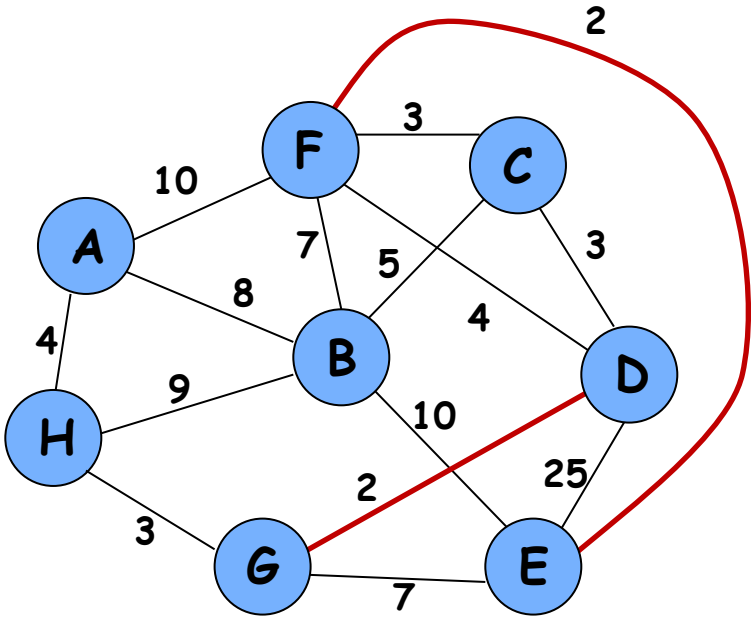
Walk-Through



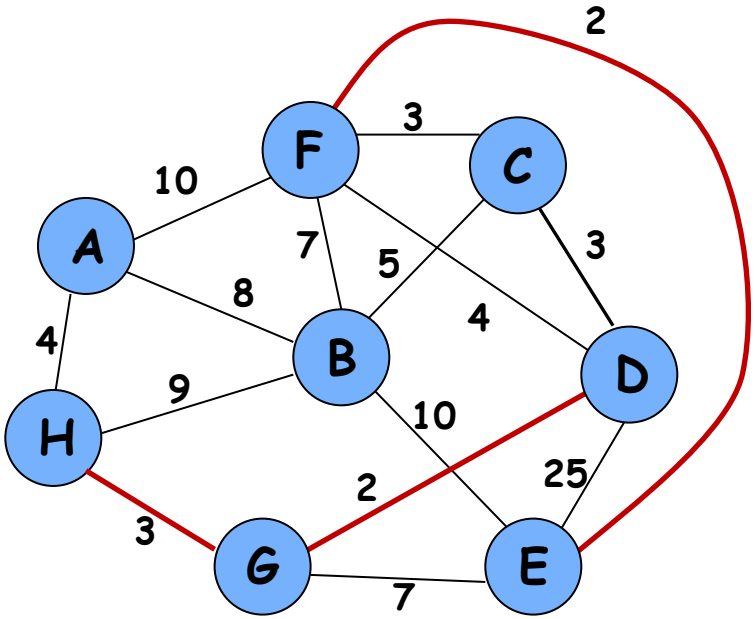
Walk-Through



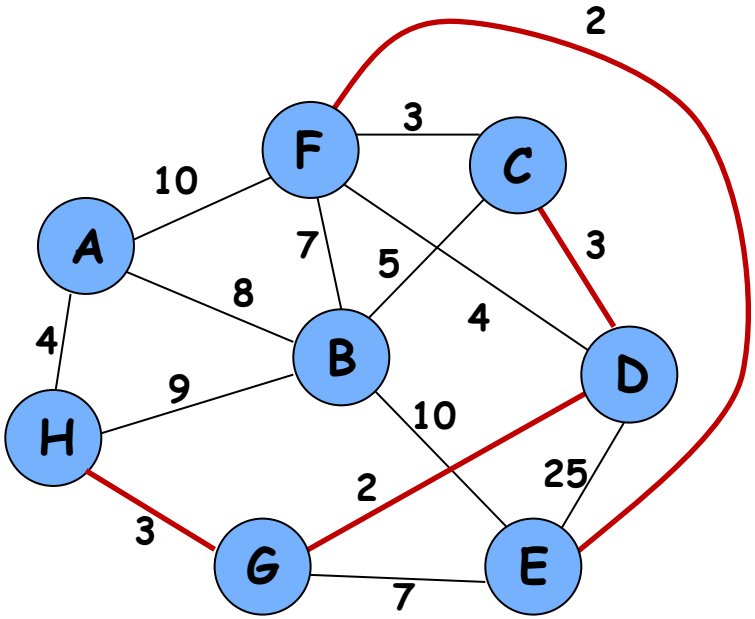
Walk-Through



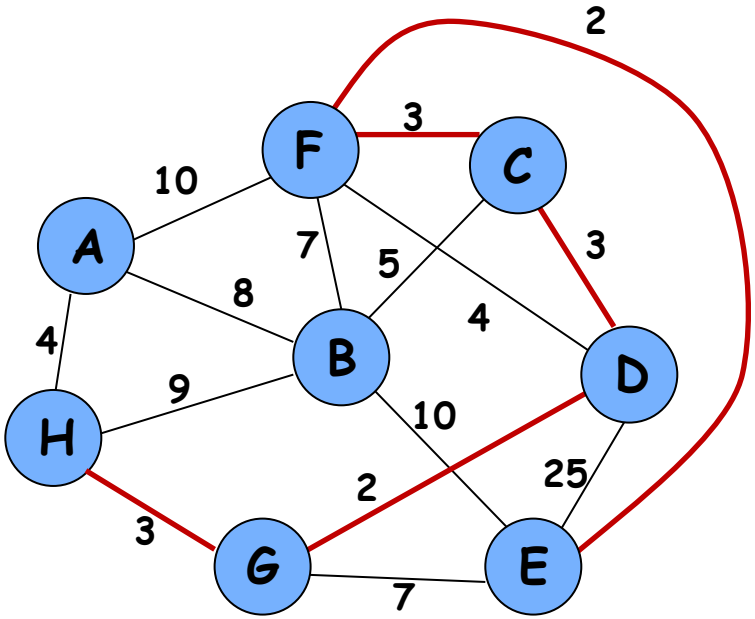
Walk-Through



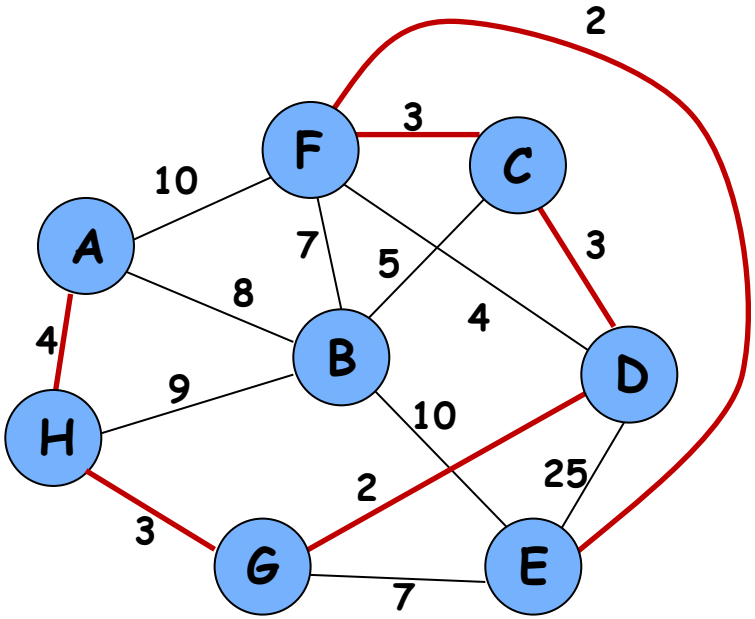
Walk-Through



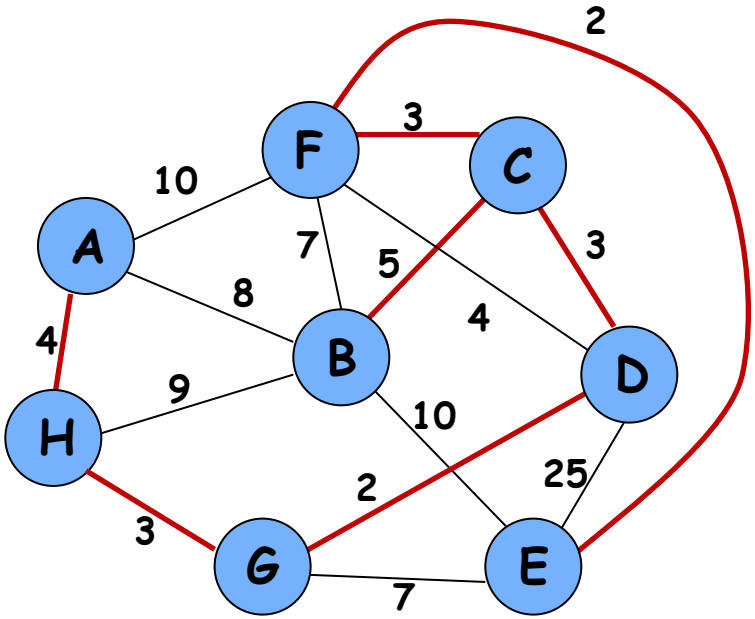
Walk-Through



Walk-Through



Walk-Through



Kruskal's Algorithm: Time complexity

Sorting edges takes $O(m \log m)$ time

We need to be able to test if adding a new edge creates a cycle, in which case we skip the edge

One option is to run DFS in each iteration to see if the number of connected components stays the same. This leads to $O(m\ n)$ time for the main loop

Can we do better?

Yes, keep track of the connected components with a data structure

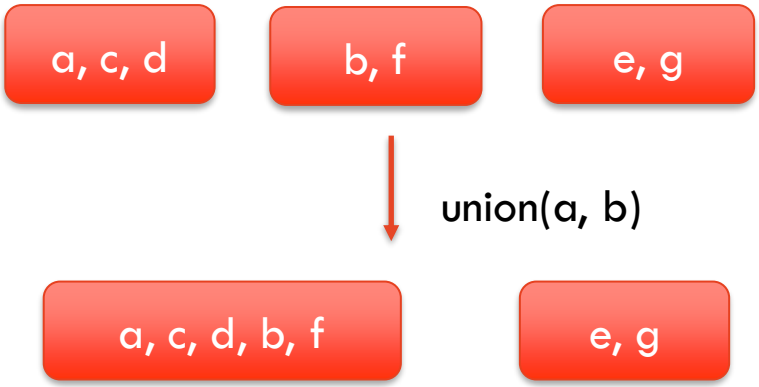
Union Find ADT

Data structure defined on a ground set of elements A

Used to keep track of an evolving partition of A

Supported operations:

- make_sets(A) : makes |A| singleton sets with elements in A
- find(a) : returns an id for the set element a belongs to
- union(a,b) : union the sets elements a and b belong to



Kurskal's algorithm implementation

```
def Kruskal(G,c):

  sort E in increasing c-value
  answer ← [ ]
  comp ← make_sets(V)
  for (u,v) in E do
    if comp.find(u) ≠ comp.find(v) then
      answer.append( (u,v) )
      comp.union(u, v)
  return answer
```

Union find operations:

- make_sets(A) : one call with $|A| = |V|$
- find(a) : 2m calls
- union(a,b) : n-1 calls

Simple union-find implementation

Sets are represented with lists. An array points to the set each element belongs to

- make_sets(A) creates and initialized the array
- find(u) is a simple lookup in the array
- union(u,v) adds elements in u's set to v's set

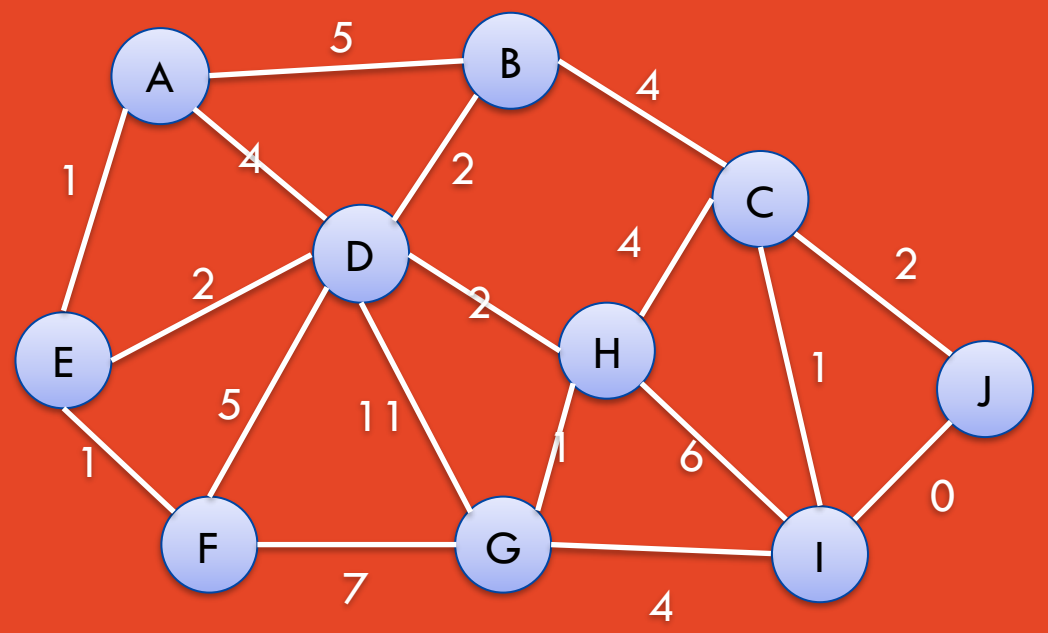
Time complexity:

- make_sets(A) takes O(n) time, where n = |A|
- find(u) takes O(1) time
- union(u,v) take O(n) time

Kruskal's algorithm would run in $O(n^2)$ time after the edge weights are sorted.

Discovery Activity: The Empire

A great battle in middle earth is taking place and each faction is aiming to grow their regional control. Each faction and their flag colour is listed below.



Elves

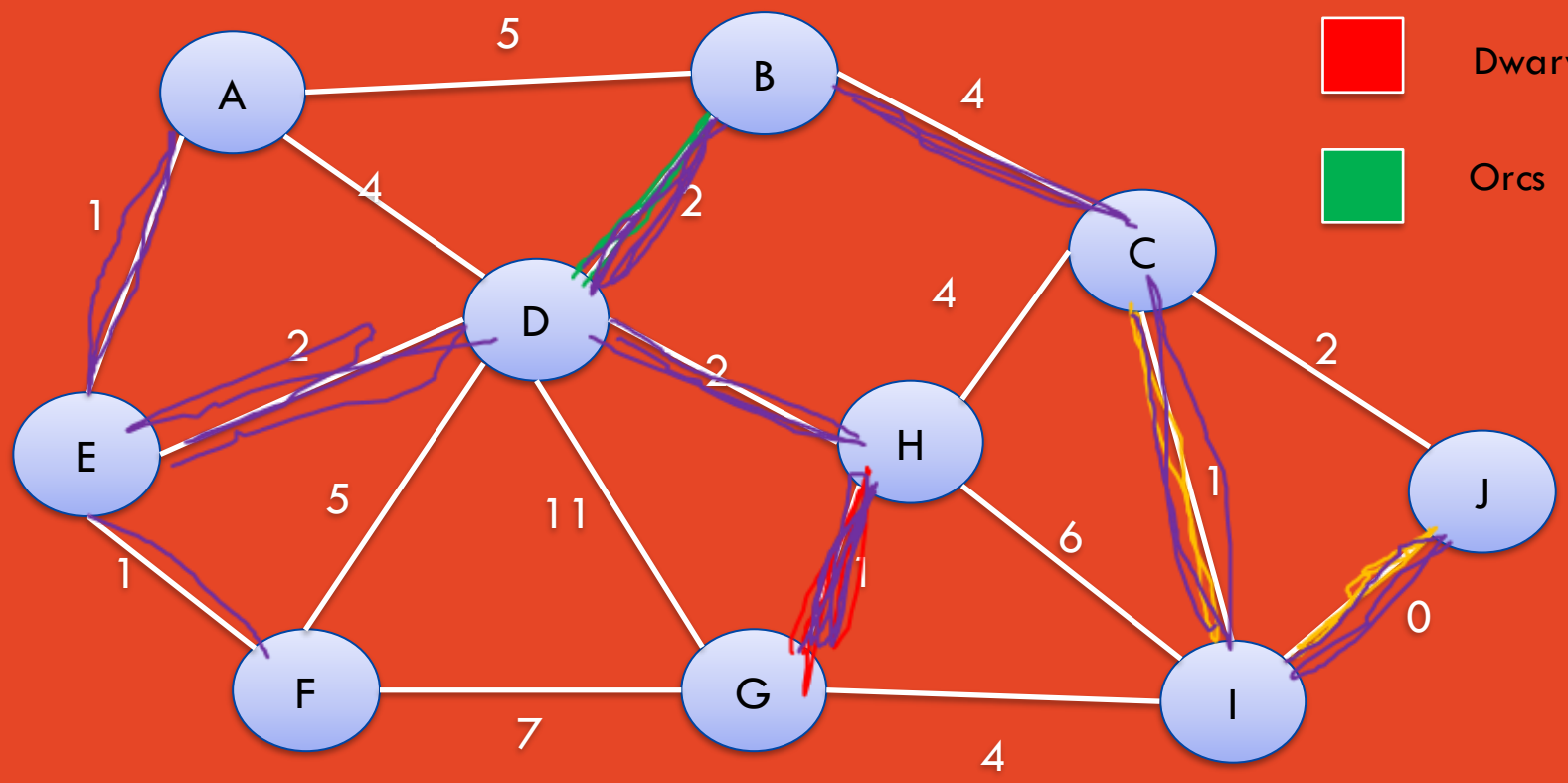
Humans

Dwarves

Orcs

1. Use the order listed above from top-down to assign your Sets in Kruskal's Algorithmand to find the MST and total edge weight.
2. Which faction will control the most land at the end of the war?

Discovery Activity



Elves

Humans

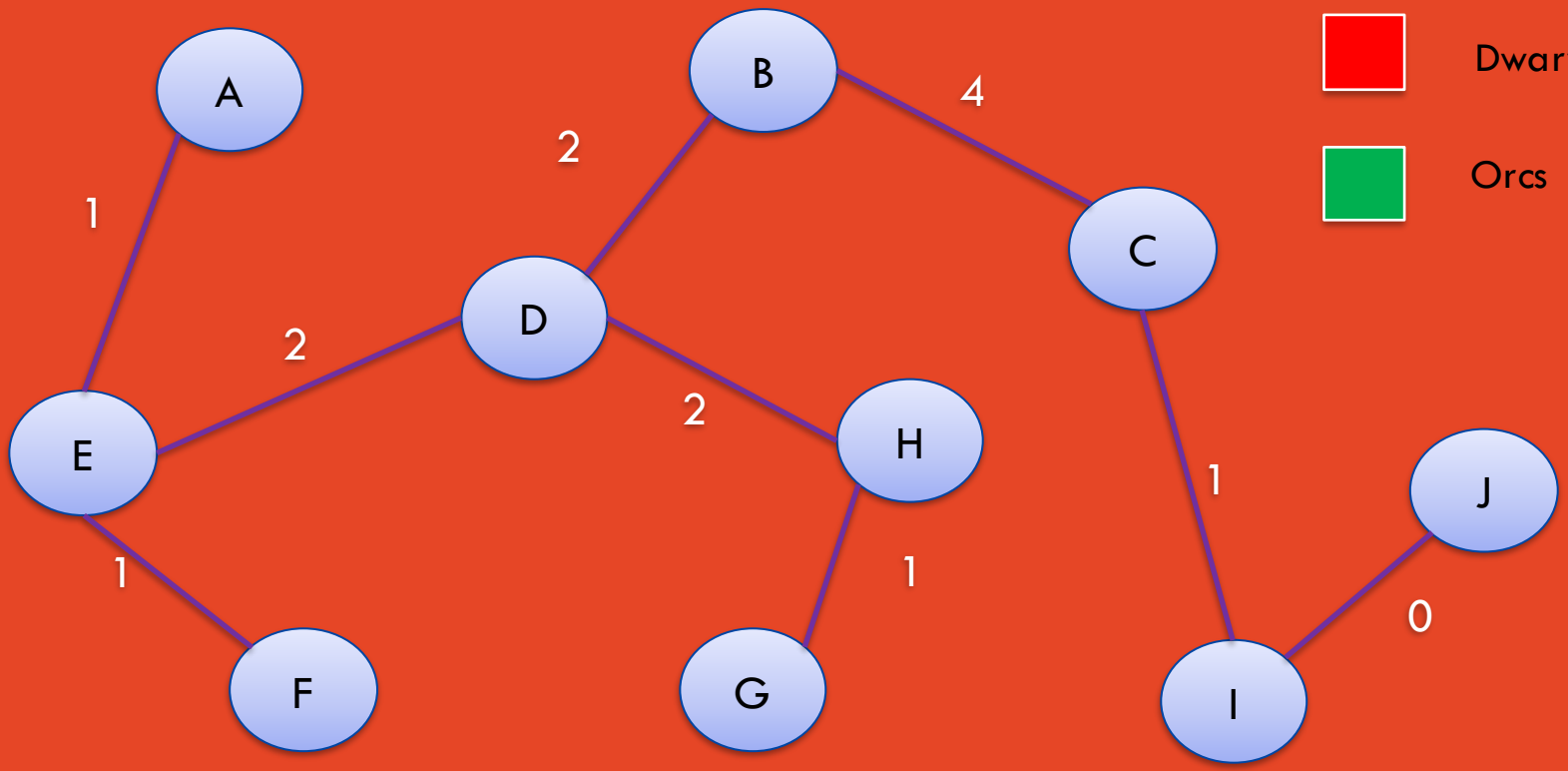
Dwarves

Orcs

I to J = 0
A to E = 1
C to I = 1
E to F = 1
G to H = 1
B to D = 2
C to J = 2
D to E = 2
D to H = 2
A to D = 4
B to C = 4
C to H = 4
G to I = 4
A to B = 5
D to F = 5
H to I = 6

1. Use the order listed above from top-down to assign your Sets in Kruskal's Algorithmand to find the MST and total edge weight.

Discovery Activity



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D to H = 2
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D to F = 5
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1. Use the order listed above from top-down to assign your Sets in Kruskal's Algorithmand to find the MST and total edge weight.